

Specification Surfaces for Incremental Predictive Performance: Estimation, Uncertainty, and Multi-Log Evaluation

Sudhir Dixit^a

^a*Independent Researcher, Apex, United States*

Abstract

What a recorded field is worth to a prediction is reported as one number, but that number is a functional of choices the report rarely states: the baseline, the pipeline, the outcome predicted, the split, the moment the decision is taken, how completely the register behind the field is populated, and the metric. We define incremental predictive performance as a *specification surface* $V_s(f)$ over a design space the analyst declares, and give three objects for reporting one: an exact functional-ANOVA decomposition of its variance that separates analyst choices from the resampling a design also contains; simultaneous max- t bands from a pipeline-refitting bootstrap, whose coverage we measure against a known answer and find short of nominal; and *resolution regions*, which say where a sign is determined. On 19 pairs from 13 public event logs, resampling carries a median 71.7% of the variance, and net of it no single analyst choice dominates on 7 pairs. Most specifications leave the sign undetermined; where the data determine one, admissible specifications sit a median 0.0430 AUC from the conventional report. In a case study on 45,455 bank incidents, a register's worth turns on the decision time and its sign on the target.

Keywords: specification surface, incremental predictive performance, predictive process monitoring, simultaneous inference, sensitivity analysis, configuration management database

Email address: `sudhir.dixit1@gmail.com` (Sudhir Dixit)

Highlights

- A field’s incremental predictive performance is a surface, not one number
- An exact decomposition separates analyst choices from resampling
- Simultaneous bands over the whole declared family, with coverage measured
- Resolved cells sit 0.0430 AUC from the one-number report
- A register’s worth turns on the decision time and on which target is set

1. Introduction

A claim that a recorded field is worth something is a comparison. Its second arm — performance *without* the field — is a choice the analyst makes, and so are the learner that makes the comparison, the encoding it uses, the outcome it predicts, the way the data are split in time, whether the field is available at the moment the decision is taken, how completely the register behind the field is populated, the metric the difference is read in, and the operating point at which that metric is evaluated. Each is routinely left implicit, and the result is reported as one number.

This paper’s position is not that such a number depends on those choices — that is known, and Section 2 says by whom — but that the dependence is large enough, on real decisions, that the number alone is not a report; that the right object is therefore a surface over a *declared* design space; and that a surface admits estimation and inference of its own. Two things follow that a reader may not expect, and both are stated here before anything is argued for. **A pooled specification surface overstates the case:** most of its variance here is resampling rather than analyst latitude, and the correction is exact (Section 4.5). And **the sign is the wrong thing to count:** where the data determine a sign, the conventional report is usually right about it and usually wrong about the magnitude, which is what Section 6.4 reports instead.

The estimand. Write a specification s for a complete assignment to every design axis, and

$$V_s(f) = m_\theta(A_s(D_{\text{train}}, B \cup \{f_q\}), D_{\text{test}}) - m_\theta(A_s(D_{\text{train}}, B), D_{\text{test}}), \quad (1)$$

where A_s is the fitting procedure (learner, encoding, hyperparameters, calibration), B the baseline feature set, f_q the field under a register quality mechanism q , m the metric and θ the operating point, and where $D_{\text{train}}, D_{\text{test}}$ are fixed by the split rule and the cohort. The *specification surface* is the map $s \mapsto V_s(f)$ over a design space $\mathcal{S}$ that the analyst declares and the reader can enumerate; Figure 1 draws one.

We say *incremental predictive performance* and not *value*: (1) measures how much better a model predicts, not what the field costs to acquire or maintain, whether acting on the prediction changes the outcome, or what an organisation would pay for either. Section 8 converts the increment into net benefit at a declared exchange rate, which is as close to economic value as this evidence gets.

Contributions.

1. **An estimand, a declared surface and an exact decomposition** (Table 1). The surface over a design space the reader can enumerate; three *denominators* distinguished and audited from the files, so that a claim’s quantifier is a counted set rather than a phrase; and an exact functional-ANOVA decomposition of the surface’s variance in which the split is an *error stratum*, so that resampling is separated from the choices an analyst makes (Section 4.5).
2. **Simultaneous inference whose coverage is measured, and the labels that rest on it.** Simultaneous max- t bands over the whole family a claim ranges over, from a pipeline-refitting bootstrap; *resolution regions* with a robustness index ρ under a declared minimum resolved share; and, in Section 10.4, the band’s family-wise coverage measured against a known answer — which is short of nominal everywhere in this corpus’s configuration, so the labels are reported as descriptive diagnostics and not as guarantees.
3. **A registered multi-log benchmark and a register at three decision times.** The surface is computed through one frozen pipeline on every log–target pair a pre-registered protocol admits — 19 of them, not only the ones where the answer resolves — and every corpus statistic is reported on all 19 and on the 8 where the register is a maintained one (Section S10.16). Section 7 then shows that a decision time fixes which cases are in scope, what is known about them and which fields are admissible at once, reconciles the two values the case study’s log has been reported to give, and finds the difference is the *target*: over

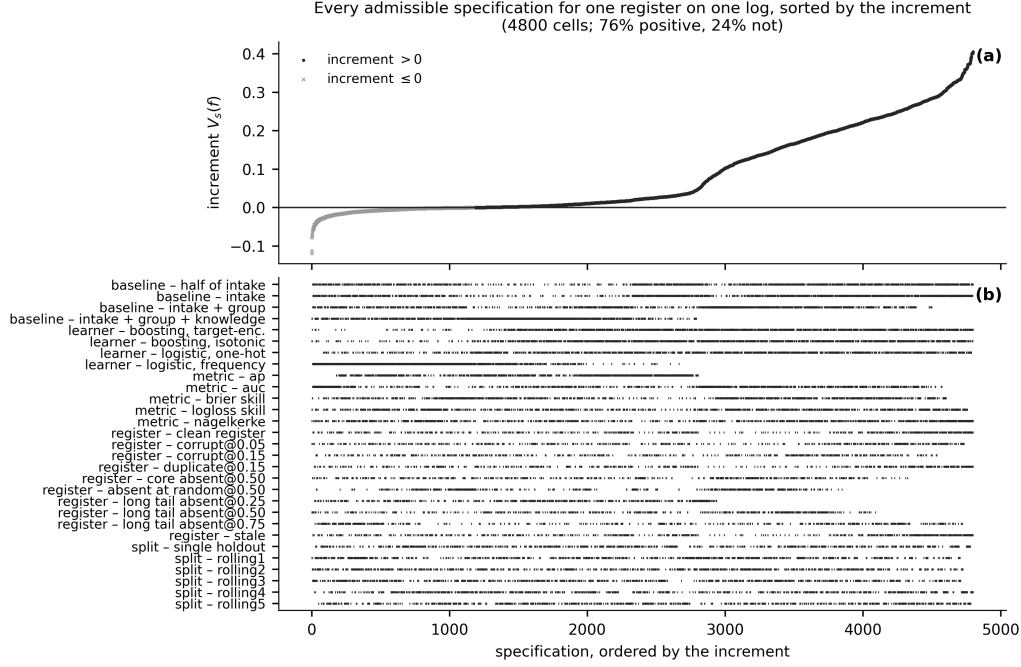
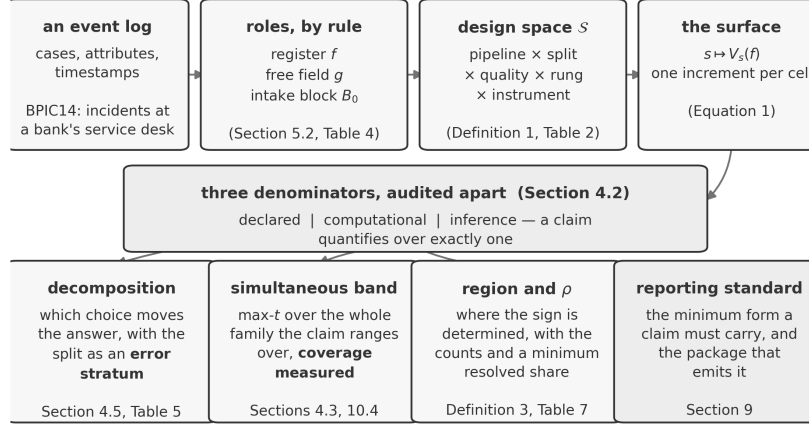


Figure 1: The specification curve for the configuration item on BPIC14, at the registered handover target: 4,800 admissible scalar cells, 4 pipelines $\times$ 4 baseline runs $\times$ 5 instruments $\times$ 10 register-quality conditions $\times$ 6 splits. **(a)** Each point is one admissible specification — a complete assignment to those five axes — sorted by the increment. **(b)** Which level of which axis each specification occupies. The intercept-only rung is excluded. Every point is a point estimate; the simultaneous bands that decide the region labels are in Table S6. **The five scalar instruments are plotted on one axis and their increments are not on a common scale** (Section 4.7); what the panel supports is the *sign* statement, which is unit-free, and not a comparison of magnitudes across instruments. The panel shows the increment rather than the reduction, so a cell where the register is worth almost nothing looks the same as one where the free field absorbs almost everything.

the cohort $\times$ target design the target carries 99.3% of the variance and the cohort 0.0%.

Specification regret — what choosing one specification costs against the best, on a scale a decision can use — is reported in full and is *not* among the contributions above. Sections 6.6 and 8.4 report that the one-number rule’s median excess regret is zero in every instrument, that the out-of-sample ordering reverses under rolling folds, and that the rules mostly tie in net benefit. It is carried as a *robustness check* on the other two arguments (Section 4.7):



Specification regret — what one specification costs against the best — is computed from the same surface and reported as a robustness check (Section 4.7).

Figure 2: The method, once. A log is read into three *roles* by declared rules; the roles are crossed over a declared *design space*; the result is a *surface*, one increment per cell. Every claim about that surface names which of three *denominators* it quantifies over, and the reporting objects are computed from it. The schematic carries no result: the numbers are in the tables it names.

an object that reports a null is a check and not a result.

What this paper is not. This is a retrospective study on public benchmarks, offered as an evaluation methodology. It is not a procurement conclusion about configuration management databases, not a systematic review of reporting practice, and not a report of a deployed system; Section 11 says what each of those would have required.

The supplementary document. The protocol and its amendments, the proofs, the constructions of the reporting objects, the correction register, the results this article summarises and the tables it does not print are in a supplementary document whose sections, tables and figures carry an S. A reference here to Section S3 or Table S12 is a reference to that document. **The article states every claim it makes and prints the evidence for each:** which attribute plays the register on every pair and how many levels each axis carries (Table 4), the master table (Table 5), what a region label rests on (Table 7), the decomposition net of resampling (Table 6), the decision-time ladder (Table 9) and the band’s measured coverage (Table S37). The supplement carries the constructions, the proofs and the sensitivity analyses; a reader who wants to check an inference rather than read one will want it.

object	what it is for	§	table	what it costs
Sensitivity decomposition	how much of the disagreement each analysis choice explains, and how much belongs to no single one	4.5	S5	exact; no refit. A Möbius sum over the surface already computed
Simultaneous band	an interval that holds over every cell a claim quantifies over, not one cell at a time	4.3	S6	the whole cost: one draw refits every arm of every cell
Resolution region and ρ	which way the surface points where the data settle it, and how much of it they settle	4.6	7	a function of the band; no further refit
Specification regret (<i>a check, not an object</i>)	what choosing one specification costs against the best one, on a scale a decision can use — it returns a null on this corpus	4.7	S15	exact; no refit, on the surface already computed

Table 1: THE REPORTING OBJECTS. The three this paper supplies and the fourth it retains as a robustness check, what each is for, where it is defined, where it is reported on this corpus, and what it costs; the two middle columns are the section it is defined in and the table that reports it. Only the second costs model fits: a draw of the bootstrap refits every arm of every cell, which is why the surface an inference is taken over is smaller than the surface a point estimate is taken over (Section 4.2). The other three are functions of a surface that has already been computed, so an analyst who has done the sweep has already paid for them.

term	what it means	on the case study’s log
register f	the recorded field whose worth is in question: high-cardinality, maintained, reused across cases	the affected configuration item
free field g	a resource-like field that is recorded anyway and that the register has to beat	the assignment group
intake block B_0	what is known when the case arrives, excluding the register and the free field	category, impact, urgency, priority
rung	one baseline in the nested ladder $B_0 \subset B_1 \subset \dots$ the increment is measured against	intake + group + knowledge reference
instrument	the scalar metric an increment is read in; five are declared and they are not on a common scale	ROC AUC
cell	one complete assignment to every varied axis: a point of the surface	one-hot logistic, holdout, clean register, intake + group, AUC
specification surface	the map from cells to increments over a declared design space	the 4,800 cells of Figure 1
declared / computational / inference surface	what is admissible, what was computed, and what carries bootstrap draws — three denominators, audited apart	29,040, the same, and 3,900
resolved	the cell’s simultaneous band excludes zero, so the data determine the sign of that cell’s increment	197 of 5,808 AUC cells corpus-wide
family $\mathcal{F}$	the set of cells a claim quantifies over, which is what a simultaneous band must cover	a median 180 cells per pair
resolution region	the label a surface earns from its resolved cells: uniformly or conditionally beneficial or harmful, sign-changing, unresolved	sign-changing
robustness index ρ	(beneficial – harmful) / $ \mathcal{F} $, always printed with the counts behind it	see Table 7
MPID	the minimal practically important difference, declared before any disagreement is counted	0.01 AUC

Table 2: THE VOCABULARY, IN ONE PLACE. Every term this paper uses in a sense a reader could not guess, with the case study’s own value for it. The terms are defined again where they are first used; this table exists so that a reader who meets one in Section 6 does not have to find that place.

2. Related work

Every component used here is established. That a feature’s apparent importance depends on what else is in the model motivates conditional and permutation importance [39, 14], Shapley attributions [24, 8] and nonparametric formulations of importance as a population quantity [49]; the clinical-prediction literature has said for two decades that the increment in AUC from adding a predictor depends on the existing model [28, 7], is a poor guide to clinical usefulness [7, 44] and is routinely misread [21]; the same argument is made about effort-aware metrics in defect prediction [4], where which metric is chosen changes which method wins. Decision-analytic evaluation supplies the exchange rate [46, 47, 45]. Multiverse and specification-curve analysis supply the sweep [37, 35, 27], underspecification and variance studies the observation that pipelines differ [9, 2], and data valuation a different question about a datum’s worth [15, 18]. On event logs, outcome-oriented predictive monitoring supplies the task [41, 43, 33], encoding studies one axis [23], and data-quality work the vocabulary for a degraded register [40, 26, 6]. Leakage discipline is standard [20, 19, 48].

Four further strands bear on the argument. *Many-analysts studies* put one question and one dataset to independent teams and observe the spread directly [34, 3]: they establish that analytic variability is large without supplying an estimand for it, which is what the surface is. *Predictive multiplicity* and Rashomon sets study near-equally accurate models that disagree on individuals [25, 32] — multiplicity at the level of a fitted model, where this paper’s is at the level of the design that produced it. *Benchmark-variance work in predictive process monitoring* reports how far results move with the pipeline and the protocol [29], which is the evidence this paper’s decomposition partitions. And the *sensitivity-analysis* tradition of bounding an unmeasured assumption [42] is the form Section 7.3’s leakage tipping point takes. On the estate side, the accuracy of a configuration management database is itself a studied quantity [40, 26], and the layered service models practitioners maintain are why Section 7.4 asks which *layer* of a register pays. Supplement S1.1 sets out each strand and what this paper takes from it.

The novelty gap, precisely. Every component used here is established and none is claimed. What none of them supplies is the object this paper reports, and the gap is in four places. *The estimand is not declared completely:* the axes are discussed one at a time, and the decision time and the register’s

quality as preliminaries rather than as arguments. *The measure is left implicit*: a specification curve is summarised uniformly over whatever cells were enumerated, a choice that moves the answer (Section 4.5). *Inference is pointwise or absent*: a curve is a distribution of point estimates and a permutation test on its median, which licenses no statement about a sub-region. *Nothing measures what the conventional report costs*: the alternative to one number is offered as a display, not as a decision rule with a loss.

What is new here is narrower than the objects’ names suggest, and is stated precisely. The decomposition is an adaptation of functional ANOVA over a configuration space [17]. *Counting the share of specifications with an effect in a declared direction is already standard specification-curve practice* [35]: the robustness index ρ is that count, and this paper does not claim it. What is new is what the count is taken over and under — a *simultaneous* band over a declared family rather than a pointwise test, an *audited* denominator rather than the cells that happened to be enumerated, and a minimum resolved share below which no direction is reported at all. The decomposition with the split as an error stratum, and the separation of analyst latitude from resampling that it makes exact, we have not seen elsewhere. Section 6.5 measures where a curve’s verdict and a region label part company rather than asserting a difference.

The contribution is deliberately *not* “people usually report one number”. It is that a claim of this form is a functional of a design, that the functional can be estimated and its uncertainty quantified, and that on this corpus it is far from constant.

3. The specification surface

3.1. The design space

Definition 1 (Design space). *A design space $\mathcal{S}$ for a feature-value claim is a finite set of specifications, each a complete assignment to the axes*

$$s = \left(\underbrace{A}_{\text{learner}}, \underbrace{E}_{\text{encoding}}, \underbrace{Y}_{\text{target}}, \underbrace{\Pi}_{\text{split}}, \underbrace{\tau}_{\text{decision time}}, \right. \\ \left. \underbrace{q}_{\text{register quality}}, \underbrace{B}_{\text{baseline}}, \underbrace{m}_{\text{metric}}, \underbrace{\theta}_{\text{operating point}} \right),$$

together with the cohort and eligibility rule that fix D . A claim about a feature is well posed only relative to a declared $\mathcal{S}$.

axis	levels	measure	exclusions
learner	2-4	equal / ref. / conc.	none
encoding	2-3	set by the learner	none
split	6	equal / ref. / conc.	none
decision time (availability)	1-2	never averaged	two times on the primary log, one elsewhere
register quality x severity	6-10	equal / ref. / conc.	none
baseline (ladder rung)	4-5	equal / ref. / conc.	intercept-only: in the surface, out of every admissible set
baseline, admissible only	3-4	equal / ref. / conc.	none
scalar instrument	5	never averaged across	none
operating point (net benefit)	31	declared threshold dist.	outside the declared range

Table 3: The declared design space. One row per axis: the number of levels (a range where it differs by log), the measure the paper places on those levels, and every level excluded by declaration with its reason. Generated from the code that walks the space, so the level counts multiply to the cell counts of Table S11.

Declaring a design space means writing down every axis, its levels, the measure over those levels, and every level that is deliberately excluded with the reason. Table 3 is that declaration for this study, and it is generated from the code that walks the space rather than typed beside it. A reader can multiply its level counts and obtain the cell counts reported in Section 6; that this multiplication works is the first check the verification harness runs. **The nine axes above are the estimand’s arguments, and this study does not vary all of them.** The target Y indexes the surfaces rather than being an axis *of* one — each log–target pair has its own surface — and the decision time, the operating point and the cohort are varied on the case study’s log alone. Table 4 states, per pair, which axes ran and at how many levels; the two tables are meant to be read together and Section 11 carries the gap between them. Table 3’s *measure* column names which of the three declared product measures over the axis levels a quantity is taken under — EQUAL, every level of every axis equally likely; REF., half the mass on the reference level; CONC., four fifths on it — and Section 4.5 reports every decomposition under all three.

The measure placed on the axis levels is a modelling choice and is declared rather than left implicit; Supplement S1.2 states it, and says why the decision time and the register’s quality are axes of the estimand rather than preliminaries to it.

3.2. What the surface estimates

Equation (1) is written on a fixed D_{train} and D_{test} , so as it stands $V_s(f)$ is a deterministic functional of the log in hand and an interval for it would cover with probability zero or one. That is not the quantity this paper reports

intervals for, and the distinction has to be made before Section 4 can say what its bootstrap targets.

Definition 2 (The estimand). *Let the cases of a log be a sample from a population P , and let the specification s fix the fitting procedure A_s , the cohort and eligibility rule, the split rule Π , the decision time, the register-quality mechanism, the baseline and the instrument. Write $A_s(P)$ for the limit of the fitted scoring rule as the sample grows under P with s held fixed. **That limit needs an assumption, and it is stated here rather than where it is used:** the case sequence is stationary and mixing, so that a temporal split does not separate two different populations and the moving-block bootstrap of Section 4.1 resamples a dependence structure that persists. An event log is not guaranteed to have that property and this paper does not test it; Section 11 carries it as a threat. The estimand at s is*

$$\vartheta_s(f) = m_\theta(A_s(P, B \cup \{f_q\}), P) - m_\theta(A_s(P, B), P),$$

and $V_s(f)$ of (1) is its plug-in estimate on the log.

Three things follow and are used throughout. The split rule and the pipeline are *part of the estimand*, not nuisances to be averaged away: ϑ_s is what this analyst’s procedure converges to, which is why the nested bootstrap of Section 4.1 resamples the training half rather than conditioning on it, and why the split is an axis. Under misspecification ϑ_s is not what the register is worth in the world — the simulation of Section 10 separates the two and measures the gap, calling them V_{limit} and V_{oracle} . And every cell of every figure and table in this paper is an *estimate* of ϑ_s ; “resolved” in Definition 3 means that the band excludes zero for ϑ_s , not that the observed increment is nonzero.

3.3. The surface, and why it is not a scalar

Given $\mathcal{S}$ and data D , the surface is the map $s \mapsto V_s(f)$ of (1), and a conventional report is one cell of it. The paper’s claim is that a single cell is not a summary of the surface in any useful sense. **The point estimates alone establish that:** on 18 of the 19 pairs, between a tenth and nine tenths of the admissible cells are positive, so the surface’s sign is not settled by the data and the design together. Whether that unsettledness amounts to a *misstatement* is a different and much harder question, and Section 6.4 answers it with three restrictions and a withdrawal rather than with a rate.

There is one state in which a scalar is defensible: when the whole admissible surface lies on the same side of zero with simultaneous confidence, a positive number is a lower bound on a robust conclusion. Section 4.6 makes that a definition.

3.4. The relative reduction, and its pathologies

Comparisons between two baselines are often stated as a proportion. Write $B_0 \subset B_1$ and

$$D_s(f) = V_s(f | B_0) - V_s(f | B_1), \quad R_s(f) = 1 - \frac{V_s(f | B_1)}{V_s(f | B_0)} = \frac{D_s(f)}{V_s(f | B_0)}. \quad (2)$$

D is an absolute difference in the metric’s own units. R is a ratio and inherits every pathology of one: the denominator can be small, zero or negative; R is not comparable across metrics, because Remark 1 says the two are not functions of each other; and R can exceed 1 or fall below 0, so reading it as “the share absorbed” is a category error whenever it leaves $[0, 1]$.

We therefore make D and the two increments primary throughout and treat R as a secondary descriptive statistic subject to three rules, declared before any R in this paper was computed:

1. R is reported only where $V_s(f | B_0)$ is resolvably positive under the *simultaneous* band, not merely positive as a point estimate;
2. its interval is a Fieller set and its *kind* — bounded, unbounded, or the complement of an interval — is printed beside it. A confidence set for a ratio is the solution set of a quadratic [13] and is a bounded interval only when that quadratic’s leading coefficient is positive, which is to say only when the denominator is resolvably away from zero; otherwise it is unbounded or is the *complement* of an interval, and no bounded interval summarises it (Supplement S3). Of the 10,584 reductions this paper computes, 5,282 have a non-bounded Fieller set, and a percentile interval reports a bounded one for every single one of them;
3. the magnitude of R is never compared with the spread of R across metrics as though both lived on an unproblematic common scale.

3.5. Two facts about the metric, and why they are remarks

Two properties of the reduction are used later and are stated here as remarks rather than proved here; Supplement S2 carries both statements in full, their constructions, the richness assumption the second needs, and the witness that separates it from rank-basedness.

Remark 1 (The metric is an argument of the estimand). *No function carries the AUC reduction to the average-precision reduction: two configurations can share an AUC reduction to 2.2×10^{-16} while their average-precision reductions differ by 0.417. The reduction is therefore not identified by the data, the feature and the two baselines, which is why every table in this paper names its instrument and why the minimal practically important difference of Section 6.4 is declared in one instrument’s units and applied there only.*

Remark 2 (What survives a declared family of recalibrations). *For a family Ψ of strictly increasing maps applied simultaneously to all four arms of a configuration, the reduction is Ψ -invariant exactly when the metric is affine under Ψ . Rank-basedness is sufficient and not necessary, and the gap is family-relative: the class-mean gap — the difference in mean score between the two outcome classes, which is not rank-based — has a reduction invariant to 1.9×10^{-14} inside the affine family and shifting by up to 0.638 over 5 non-affine ones. What this buys is that a search becomes a proof: one map and one configuration violating the invariance rules it out everywhere, and 18 such certificates are computed, the smallest moving the reduction by 0.241, against 2,400 checks in which the two rank-based instruments hold to exactly zero.*

4. Estimation and inference

4.1. The estimator, and uncertainty that includes the model

The pipeline is frozen and identical on every log, and the *pipeline* axis spans encodings as well as model families. Attributes are assigned to roles by the pre-registered rules of Section 5, cases are ordered by their first timestamp, and the split is temporal; Supplement S3.2 lists the levels and why the tree learner is target-encoded. The usual interval for such an increment resamples test rows with the models held fixed, which excludes training-sample, model-selection, encoder and split variability and the temporal dependence between cases.

Every interval here is instead a *nested moving-block bootstrap*: one draw resamples the training half by moving blocks [22] of length $\lceil n^{1/3} \rceil$, refits every arm of every rung under every learner and quality mechanism, and evaluates on a moving-block resample of the test half, with encoders, frequency tables, target encodings, register rankings and calibrators rebuilt inside the draw. On the simulation of Section 10 the fixed-model interval covers a median

92.1% and the nested basic interval 93.0% — **a gap of about 1.3 Monte Carlo standard errors** at the 0.7% standard error that design attains (Section 10.2), which is not a difference this evidence can resolve. The nested construction is retained because the variability it admits is variability the estimand contains and the fixed-model interval excludes by construction, not because it measures better here; and the price it charges is the subject of the next two paragraphs. Section 11 records that the comparison a reader would want — against a *weighted* bootstrap, which would remove that price at source — is the first thing this design should be replaced by, and is not run here.

Throughout, K denotes the **cardinality of the register field** — the number of distinct values f takes on a pair, 337 at the median pair and 5,245 at the largest (Table 4) — and n the number of training cases. The ratio K/n governs everything in this subsection and in Section 6.3.

Refitting inside the draw breaks the percentile interval. A bootstrap resample holds about $1 - e^{-1}$ of a high-cardinality categorical’s distinct levels, so every refit sees a sparser register and returns a smaller increment; the bootstrap distribution is displaced and a percentile interval taken from it is a correct interval for the wrong quantity. On the sparse world of Section 10 it covers 37.2% against a nominal 95%, *worse* than the fixed-model interval refitting was meant to improve on. Every interval here is therefore the *basic* (pivotal) interval $[2\hat{V} - q_{1-\alpha/2}, 2\hat{V} - q_{\alpha/2}]$ [10], and simultaneous bands are recentred by the same shift (Supplement S3.7).

A pivoted interval need not contain the estimate it is built from, and on this corpus 74 cells do not. Pivoting moves the interval by twice the displacement. Where the displacement exceeds the interval’s own half-width, both endpoints land on the same side of $\hat{V}$ and the band excludes it. That is a property of a pivotal interval under bias rather than an error, but it is not what a reader expects of a band, so it is counted: 74 of the 3,900 whole-surface cells — 1.9% — on 3 of the 19 pairs, reaching 21.1% of the cells on the worst of them. The ratio of displacement to half-width has a median 0.09 across pairs and 0.45 on that pair.

It gets worse with the sample size, not better. The displacement is a bias — a resample holds about $1 - e^{-1}$ of the distinct levels however many rows it has — so at a fixed ratio it does not shrink with n while the half-width shrinks like $n^{-1/2}$. Section 10.3 locates the crossing: at a ratio of

0.125 the interval straddles its own estimate in every replicate up to 8,000 training rows and in only 1.6% of them by 50,000. The three pairs above are the corpus’s nearest it, and Section 6.3 reports what it costs the coverage calibration.

4.2. Three surfaces, and which one a claim may quantify over

A statement of the form *this surface is uniformly beneficial* is a statement about a denominator, and the denominator has to be the set the statement ranges over. Three sets are distinguished, named, and counted from the files rather than quoted (Table S11). The **declared surface** is every scientifically admissible specification: the product of the axis levels declared in Section 5. The **computational surface** is every specification actually evaluated; here it equals the declared surface — 29,040 scalar cells against 29,040 declared, with 0 duplicates and 0 missing — and the audit script asserts that identity rather than the manuscript asserting it. The **inference surface** is every specification carrying bootstrap draws; it is smaller for one reason, that a draw refits the whole pipeline and its cost is the declared surface’s multiplied by the draw count.

The rule that selects it, in full. A pair falls into one of three size classes on its row count alone, and the class fixes the levels: the pipeline axis keeps all 4 levels on the case study’s log, the two differing in model *family* on a medium log and the two differing in *encoding* on the largest two, where a boosting fit refitted in every draw would cost more than the rest of the corpus together; the split axis keeps the two levels that bracket the declared six; the register-quality axis keeps as-recorded, a population failure and a discovery failure, dropping the third on the largest logs; the ladder and the 5 instruments are kept entire. Nothing in the rule reads a result. It is *axis-complete*, so no axis is silently fixed — but it is a *corner* of the declared surface rather than a design, which is why the decomposition cannot be computed on it and why the positive-cell share differs between the two surfaces by a median 0.067; Section 11 names the resolution-IV fractional factorial that would remove both objections. It carries 3,900 scalar family members, a median 12.5% of each pair’s computational surface. Every quantity that does *not* need a bootstrap — the decomposition, the sign-disagreement rate, the regret comparison — is computed on the full declared surface; only the region labels are computed on the inference surface.

4.3. Simultaneous bands over the family the claim ranges over

A decision curve read across 31 thresholds, a ladder read across rungs, or a surface read across every admissible cell, is a family of statements, and a pointwise interval does not control the probability that *some* member of the family is wrong. For a declared family $\mathcal{F}$ of cells with bootstrap draws $V_c^{(b)}$ and per-cell standard errors $\hat{\sigma}_c$, let

$$T^{(b)} = \max_{c \in \mathcal{F}} \frac{|V_c^{(b)} - \bar{V}_c|}{\hat{\sigma}_c}, \quad q_{1-\alpha} = \text{the } (1 - \alpha) \text{ quantile of } \{T^{(b)}\},$$

and report $\hat{V}_c \pm q_{1-\alpha} \hat{\sigma}_c$. This is the max- t (Romano–Wolf) construction [30] and has family-wise coverage over $\mathcal{F}$ asymptotically, and over nothing larger.

The family must be the family. Within-cell bands, each covering five members, do not cover the several hundred cells a surface-level label ranges over. Four interval objects are declared and every table says which it prints: POINTWISE, WITHIN-INSTRUMENT, WHOLE-SURFACE — max- t over *every* admissible scalar cell of the pair, the only family a surface-level claim may use — and DECISION-CURVE, declared separately because net benefit and a scalar instrument are not the same scale. The whole-surface family has a median 180 members against five, and its median critical value is 3.33 against 2.35 and the pointwise 1.96. Family control is per log-target pair; every cross-pair statement here is descriptive. Supplement S3.3 gives these four *interval* objects — which are not the reporting objects of Table 1 — and the rule for dropping a replicate that leaves a cell without outcome variation.

Estimating the critical value without paying for it in refits. $q_{1-\alpha}$ is the upper quantile of a maximum over hundreds of correlated cells and the empirical quantile of B observed maxima estimates it badly, while raising B multiplies the cost of the whole design space. Standardising the draws and multiplying them by independent standard normals gives a statistic with their empirical covariance, generated 20,000 times for the cost of matrix arithmetic [5]; Supplement S3 gives the construction. The budget therefore fixes B and not the precision of q — order-statistic widths of 2.01 against 0.037 — and the residual Monte Carlo interval is carried through, so a cell counts as *resolved* only at its *upper* end. A max- t band also inherits the coverage of the interval it is built from, which Section 10.3 shows is governed by the ratio of register cardinality to training size; Section 6.3 calibrates q at each pair’s own ratio.

What these bands cover, measured against a known answer. They do not attain their nominal level anywhere in this corpus’s configuration. Section 10.4 measures family-wise coverage over synthetic families matched to this corpus in size, draw count, cross-cell correlation and tails, whose truth is known by construction and whose bootstrap is *ideal*, so every coverage it reports is an upper bound on what the real procedure attains. The multiplier band’s family-wise coverage there has a median 84.6% and a minimum 76.1% against a nominal 95% — 87.3% and 74.8% even under the Gaussian draws the construction is derived for. The binding constraint is the draw count and not the choice among estimators: at the modal surface family coverage runs 82.0% at 33 draws and 95.8% at 400, and this corpus runs at 40 to 150 as declared, 33 to 150 effectively. Section 10.4 reports all five candidate critical values, the safeguard that buys almost nothing, and the shortfall in the units the calibration works in. **Every band, region label and ρ in this paper is therefore an anti-conservative statement:** they are reported as descriptive diagnostics, they are not coverage guarantees, and Section 11 states what follows for each conclusion that rests on one.

4.4. *Planned contrasts, and a p-value the design can produce*

Five contrasts on the primary log are *planned for this analysis round* — that the register adds to the intake baseline, that it still adds once the free field is admitted, that it adds at the later decision time, that the absorption D is positive, and that a half-empty register is worth less than a full one — each stated with its complete estimand in Table S12. **They are not confirmatory:** they were fixed before this analysis round, not before the data were seen, so the Holm-adjusted one-sided bootstrap p -values [16] they carry are reported for legibility and not as a confirmatory test, and 5 of them survive at $\alpha = 0.05$. Two details matter and are in Supplement S3.4: at 2,000 draws the smallest attainable Holm-adjusted value is 0.00250, because the plus-one estimator $\hat{p} = (k + 1)/(B + 1)$ is the only one a finite bootstrap can support; and on PC3 the one-sided p -value and the two-sided basic interval disagree, where **a contrast is called resolved only when the interval says so.** Every other comparison in this paper is *exploratory* and carries no multiplicity correction, because none would be meaningful over a surface enumerated in full.

4.5. The sensitivity decomposition

Because the surface is a complete factorial over its axes — the audit of Section 4.2 establishes that it is, cell for cell — the variance of V_s admits an exact and *orthogonal* functional-ANOVA decomposition [36, 31] under a product measure $\mathbb{P} = \otimes_i w_i$. Every subset u of axes has a component g_u , a Möbius sum over the conditional means computed exactly rather than estimated (Supplement S3); the components are orthogonal, so $\sum_{u \neq \emptyset} D_u = \text{Var}(V)$ exactly, and with $S_u = D_u / \text{Var}(V)$ the first-order and total indices are $S_i = S_{\{i\}}$ and $S_{T_i} = \sum_{u \ni i} S_u$. The closest precedent is Hutter et al. [17], who decompose an algorithm’s performance over its *configuration* space by the same device; what differs here is that the axes are analysis decisions rather than hyperparameters, that the components are exact over a complete factorial rather than estimated from a surrogate over a sampled design, and that the decomposition is one of three reporting objects rather than the conclusion.

The split is an error stratum, and the primary decomposition treats it as one. Five of the split axis’s six levels are expanding-origin folds on the same data. They are different *samples*, not different analyses, and a decomposition that pools them reports sampling variability as specification sensitivity. Because the decomposition is exact and orthogonal, the repair is a partition rather than a model: every component either involves the split or does not, so

$$1 = \underbrace{\sum_{\emptyset \neq u \subseteq \mathcal{A}} S_u}_{\text{analyst choice}} + \underbrace{S_{\{\text{quality}\}}}_{\text{counterfactual}} + \underbrace{\sum_{u \not\ni \text{split}} S_u - (\dots)}_{\text{mixed}} + \underbrace{\sum_{u \ni \text{split}} S_u}_{\text{resampling}}, \quad (3)$$

where $\mathcal{A} = \{\text{pipeline}, \text{rung}\}$ and the third term is what the split-free components carry beyond the first two, and the four terms are computed, not estimated (Table 6); the identity holds to 4.4×10^{-16} . The first three are what survives fold-averaging, and they are decomposed again *on* the fold-averaged surface, where each index is a share of the variance that fold-averaging leaves and the higher-order share is net of resampling. **Section 6.2 reports that decomposition as the primary one** and the pooled decomposition beside it, because the pooled one is what a reader of the earlier specification-curve literature will expect and the difference between them is itself a result.

The per-axis difference $S_{T_i} - S_i$ is the variance axis i is *involved* in beyond its main effect. These overlap — a two-way component appears in both its axes’ total effects — so their largest member is not the share carried by

interactions. That share is

$$S_{\text{int}} = 1 - \sum_i S_i, \quad (4)$$

which this paper computes together with every disjoint component, so the identity $\sum_u S_u = 1$ is asserted numerically — it holds on all 304 decompositions, which is the 19 pairs crossed with the 5 instruments under each of the three declared product measures, plus one per pair on the pooled cross-instrument scale — rather than assumed. The other quantity is retained as the *largest per-axis interaction involvement*; the two differ substantially, at 43.7% against 56.0% at the median pair.

4.6. Resolution regions and the robustness index

Definition 3 (Cell labels and surface regions). *Let $\mathcal{F}$ be the declared inference family of Section 4.2 for a pair — the cells that carry bootstrap draws, a median 12.5% of that pair’s admissible scalar cells. Under the simultaneous band of Section 4.3, a cell of $\mathcal{F}$ is beneficial if its lower band is above zero, harmful if its upper band is below zero, and unresolved otherwise. A surface is uniformly beneficial if every cell of $\mathcal{F}$ is beneficial; conditionally beneficial if some are and none is harmful; uniformly harmful and conditionally harmful in mirror image; sign-changing if there is at least one cell of each kind; and unresolved if none is resolved — **or if fewer than 5% of $\mathcal{F}$ is resolved at all**. The robustness index is*

$$\rho = \frac{\#\{\text{beneficial}\} - \#\{\text{harmful}\}}{|\mathcal{F}|} \in [-1, 1],$$

and every directional label is reported with the counts of beneficial, harmful and unresolved cells it rests on.

The minimum resolved share is not free. Without it a surface earns a direction from as few as two resolved cells of a hundred and eighty: *conditionally beneficial* then means “nothing resolved harmful”, which on a family the data barely touch is a statement about the band’s width and not about the register. 4 of the 13 pairs that carry a direction lose it under the condition (Table 7), and ρ is printed beside the triple everywhere because ρ alone cannot distinguish two resolved cells from ninety. The threshold is a declared convention; the labels at 0 and 6 per cent are in Table 7’s source file, so a reader who prefers another can see what it costs.

The label is relative to $\mathcal{F}$ and not to the whole admissible set, and the direction of that is not neutral. A label computed on a sixth of a pair’s cells is systematically cleaner than one computed on all of them, and *uniformly beneficial* is the label most helped by it: a claim about every cell is easier to sustain over fewer cells. The manuscript therefore never says a surface is uniformly beneficial without this restriction being in force, and Section 6.3 reports that the label is not reached on any pair in any case. The two harmful labels are not decoration: Table S6 carries 1 conditionally harmful and 0 uniformly harmful surfaces out of 19. $\rho = 1$ is the only state in which one positive number is a safe summary of a surface, and the empirical question this paper answers is how often it obtains. The region is defined over the *scalar* family; folding the decision-curve cells into it would let a single extreme threshold decide the label of a whole surface, so the decision curve gets its own region (Section 8). The intercept-only rung is excluded from every admissible set, because a reduction measured against a model with no features is inflated by construction; the master surface file *does* carry that rung, which is why the two counts in Table S11 differ.

4.7. *Specification regret, as a robustness check*

This subsection defines a check and not a reporting object. What *specification regret* reports is a null — the one-number rule’s median excess regret is zero in every instrument, the out-of-sample ordering reverses under rolling folds (Section 6.6), and on calibrated net benefit the rules mostly tie (Section 8.4) — and a null is a check on the paper’s other two arguments rather than a result of its own. It is reported in full, and it is not a contribution.

The construction asks what choosing one specification costs against the best one, on a scale a decision can use. Three designs are declared and all three reported: **A**, metric-specific, so that no average crosses an instrument; **B**, on calibrated net benefit, which is a common operational utility; and **C**, partially identified over a declared set of monotone utility maps, which reports an interval rather than assuming one map. The identity map’s maximum excess is exactly zero. Supplement S3.1 gives the three constructions and the four decision rules compared under them.

5. The registered benchmark design

5.1. *What was fixed, and when*

The rules that turn an arbitrary event log into an instance of this problem were committed before any result on a new log existed, in `PROTOCOL.md` for the corpus and `AUDIT-PROTOCOL-2.md` for the practice pilot. The version history shows the registering commit adding one file and no result file; `git log --stat` is the check, and `REPRODUCE.md` names the commit. Six amendments were declared afterwards, each with its date, its cause and the result it changed, and all six are in Supplement S1 rather than folded silently into the rules.

This is not a timestamped third-party registration, and the difference matters. A registration on a public registry is attested by a party with no stake in the result; what is offered here is a file in a repository whose history the authors control, so a reader who does not trust that history has nothing else to check. Any claim in this paper that a rule was fixed in advance is therefore a claim about the repository and no stronger than the repository. We say so plainly instead of letting the word *registered* carry a weight it has not earned: what the design buys is that the rules are legible, dated and amendable only on the record, not that an outside party watched them being written.

5.2. *Roles*

A log is admitted if the registered rules can find three things in it without reading an outcome.

The register *f*. A high-cardinality attribute that is not resource-like, has at least 50 distinct values, is reused across at least two cases on average, and — by amendment 1 — whose values recur across the temporal split, so that a document number is not mistaken for a maintained entity.

The free field *g*. A resource-like attribute, present on at least 50% of cases, whose value varies within a case, so that it can carry information about routing. **Only the *first* value it takes on a case is used as a feature**, while the handover target is defined from the whole sequence of values — so the feature is what is known when the case opens and the target is what happens afterwards, and the field cannot inform the model about its own future. The distinction is not cosmetic: the

log	target	ITSM	register field f	free field g	axes: pipe/split/qual/rung	cases	levels of f
BPIC13_closed	handover	yes	product	org:resource	2/6/6/3	1,487	337
BPIC13_incidents	duration	yes	product	org:resource	2/6/6/3	7,554	704
BPIC13_incidents	handover	yes	product	org:resource	2/6/6/3	7,554	704
BPIC14	duration	yes	CI Name (aff)	assignment group	4/6/10/4	46,606	3,019
BPIC14	handover	yes	CI Name (aff)	assignment group	4/6/10/4	46,606	3,019
BPIC15_1	duration	–	case:parts	org:resource	2/6/6/3	1,199	89
BPIC15_1	handover	–	case:parts	org:resource	2/6/6/3	1,199	89
BPIC15_3	duration	–	case:parts	org:resource	2/6/6/3	1,409	94
BPIC15_3	handover	–	case:parts	org:resource	2/6/6/3	1,409	94
BPIC15_4	duration	–	case:parts	org:resource	2/6/6/3	1,053	53
BPIC15_4	handover	–	case:parts	org:resource	2/6/6/3	1,053	53
BPIC15_5	duration	–	case:parts	org:resource	2/6/6/3	1,156	107
BPIC17	duration	–	case: RequestedAmount	org:resource	2/6/6/3	31,509	701
BPIC19	duration	–	case:Vendor	org:resource	3/6/6/3	251,734	1,975
Helpdesk	duration	yes	customer	Resource	2/6/6/3	4,580	394
RoadFines	duration	–	article	org:resource	3/6/6/3	150,370	66
Sepsis	duration	–	Diagnose	org:group	2/6/6/3	1,050	147
UCI498	duration	yes	caller_id	assigned_to	2/6/6/3	24,918	5,245
UCI498	handover	yes	caller_id	assigned_to	2/6/6/3	24,918	5,245

Table 4: ROLES AND LEVELS, PER PAIR. Which recorded attribute plays the register f and which the free field g on each admitted pair, whether the pair is IT service management, and how many levels each varied axis carries: pipeline, split, register-quality condition, baseline rung. Every pair also carries the 5 scalar instruments, so the per-pair factorial is 5 axes wide and not the nine of Definition 1; Section 5 says which axes are fixed on which pairs and why. ‘cases’ is the admitted case count and ‘levels of f ’ the register’s cardinality. This table was Supplement Table S11 through round twenty-five; the master table cannot be read without it.

same attribute supplies both, and using the case-level set would put the answer in the features.

The intake block B_0 . Every remaining attribute of cardinality at most 200 that is neither an outcome, nor a timestamp, nor a relabelling of the activity, nor a deterministic coarsening of f .

Coarsenings of f — its type, its subtype, the service it belongs to — are given their own role rather than being swept into B_0 , because which *layer* of a register pays is a separate question, answered in Section 7.4.

Table 4 is what the rules found, and it is also the design space as it was actually varied. It names, per pair, the attribute that plays each role and the number of levels each axis carries. Two things are visible there that Definition 1 and Table 3 do not say. The *target* is not an axis of any surface:

the surface is computed per log–target pair, so each of the two targets a log carries has a surface of its own, and the cohort $\times$ target design exists only in Section 7.1. And the per-pair factorial is 5 axes wide — pipeline, split, register quality, baseline rung, instrument — not the nine of Definition 1: the decision time varies only on the case study’s log, the operating point only in the decision-curve family, and the cohort only in Section 7.1. Definition 1 declares what the estimand’s arguments *are*; Table 4 declares which of them this study *varied*, and Section 11 carries the difference as a limitation rather than leaving a reader to find it.

5.3. Targets, splits and exclusions

Two targets are defined for every log — *handover*, whether more than one resource group touches the case, and *duration*, whether it runs longer than the training half’s median. Cases are ordered by first timestamp and split 70/30, with five expanding-window rolling-origin folds beside it, and fields determined only at closure are excluded by name [20, 19, 48]. A pair is excluded only for a registered reason, and **every admitted pair is analysed and reported** — running the surface only where the reduction resolves would condition the analysis on the outcome. Supplement S1.3 gives the rules, the exclusion codes and the admission margin at the upper edge of the prevalence window.

Two exclusions a reader of Table 4 will ask about. BPIC15_2 is excluded under SMALL_N at 832 cases while its four siblings are admitted — the rule is a floor on the case count and nothing about the log’s content — and the further BPIC15 targets, the BPIC20 logs and BPIC12 are excluded under NO_HEADROOM or NO_G, which are prevalence and role failures. **And the registered handover target on the case study’s log sits at the upper edge of the admission window:** test prevalence 0.927 against a window that closes at 0.95, which excludes other logs at 0.959. That pair carries 4,800 of the corpus’s 29,040 admissible cells — the log carries 9,600 across its two targets — so a corpus statistic pooled over cells is partly a statement about it; Section S10.16 reports every headline on the ITSM family separately for this reason, and Section 11 asks whether the window should be tightened.

5.4. What the corpus does and does not have in common

The corpus is 13 public logs spanning 6 domains, and it includes the further ITSM logs the literature uses [38, 1]. It is heterogeneous and the paper does not claim otherwise. The registers are a configuration item, a

caller, a product, a vendor, a diagnosis and a legal article; the free fields are an assignment group, an assignee, a resource and a role; the targets mean different things in a bank’s incident queue, a hospital’s admissions and a municipality’s permits.

Table S10 makes that explicit rather than leaving it to a sentence. For every pair it names the domain, the field playing the register role, why that field behaves as a maintained and reused entity, the free field, the interpretation of the target, and the specific threat to construct validity that the pair carries. A diagnosis code is admitted as a register by a cardinality-and-reuse rule, and it is not a configuration management database; a legal article is reused across cases but nobody maintains a discovery process for it. Those are threats, they are named per pair, and the paper’s conclusions are stated at a level that survives them.

This is therefore a *benchmark of specification sensitivity across heterogeneous prediction problems*, not a replication of one finding on several organisations, and two disciplines follow. No domain-specific effect magnitude is averaged across domains as though it estimated one universal register value: the corpus-level statistics are all about *sensitivity*, which is a property each pair has separately and which can be summarised across pairs without assuming they measure one thing. And the 8 ITSM-like pairs, the closest family to the case study, are analysed as a group as well, with *every* headline corpus statistic reported on both populations in Section S10.16.

6. Multi-log results

Every point estimate in this section comes from one master surface — one file, named in the code-availability statement — computed by one pass of one pipeline over 19 log–target pairs, 4 learners, 6 splits, 10 register-quality conditions, the nested baseline ladder and 36 instruments including the 31-point decision-curve grid. Its 273,024 rows *include* the intercept-only rung, so that the surface is complete; no admissible set contains it, and the 29,040 admissible scalar cells every headline is computed on exclude it. Every interval is the basic construction of Section 4.1 and every band the whole-surface simultaneous construction of Section 4.3, and each table names which it prints. The denominator audit is in Table S11: declared equals computed on all 19 pairs, with 0 duplicates and 0 missing, and the inference surface (3,900 scalar members) is a subset of it cell for cell, which `verify_numbers` asserts.

log	target	V at reference	pointwise	simultaneous	region	resolved share	rho	misreport rate
BPIC13_closed	handover	-0.009	[-0.041, +0.018]	[-0.068, +0.046]	unresolved	0.0	0.000	0.284
BPIC13_incidents	duration	0.028	[+0.013, +0.060]	[-0.003, +0.074]	unresolved (below minimum share)	1.7	0.017	0.393
BPIC13_incidents	handover	0.037	[+0.030, +0.050]	[+0.021, +0.059]	conditionally beneficial	42.2	0.422	0.111
BPIC14	duration	0.096	[+0.089, +0.109]	[+0.081, +0.119]	conditionally beneficial	69.2	0.692	0.106
BPIC14	handover	0.167	[+0.143, +0.205]	[+0.115, +0.228]	sign-changing	56.0	0.552	0.244
BPIC15_1	duration	0.001	[-0.029, +0.046]	[-0.073, +0.071]	unresolved (below minimum share)	1.1	0.011	0.423
BPIC15_1	handover	0.007	[-0.007, +0.021]	[-0.017, +0.032]	unresolved	0.0	0.000	0.242
BPIC15_3	duration	0.035	[-0.000, +0.070]	[-0.025, +0.094]	conditionally beneficial	5.6	0.056	0.191
BPIC15_3	handover	-0.003	[-0.047, +0.058]	[-0.088, +0.088]	unresolved	0.0	0.000	0.616
BPIC15_4	duration	0.035	[+0.016, +0.084]	[-0.008, +0.101]	unresolved	0.0	0.000	0.623
BPIC15_4	handover	0.001	[-0.026, +0.056]	[-0.059, +0.076]	unresolved	0.0	0.000	0.618
BPIC15_5	duration	0.032	[-0.026, +0.100]	[-0.075, +0.143]	unresolved	0.0	0.000	0.328
BPIC17	duration	0.002	[-0.003, +0.010]	[-0.008, +0.018]	conditionally beneficial	13.9	0.139	0.353
BPIC19	duration	0.024	[+0.011, +0.043]	[-0.001, +0.052]	sign-changing	31.7	-0.083	0.360
Helpdesk	duration	-0.020	[-0.039, -0.003]	[-0.054, +0.007]	unresolved (below minimum share)	1.1	-0.011	0.257
RoadFines	duration	0.031	[+0.025, +0.037]	[+0.020, +0.042]	sign-changing	52.5	0.225	0.173
Sepsis	duration	0.153	[+0.114, +0.210]	[+0.076, +0.237]	conditionally beneficial	28.3	0.283	0.056
UCI498	duration	-0.007	[-0.011, +0.006]	[-0.016, +0.015]	conditionally beneficial	9.4	0.094	0.493
UCI498	handover	-0.003	[-0.007, -0.001]	[-0.009, +0.002]	unresolved (below minimum share)	4.4	0.044	0.481

Table 5: The master table. One row per admitted log–target pair. ‘pointwise’ is the 95% interval at the reference cell, basic (pivotal) construction; ‘simultaneous’ is that same cell’s WHOLE-SURFACE simultaneous band, at the conservative end of the critical value’s Monte Carlo interval. They are different objects, and the region label uses only the second. Then the resolution region and the robustness index ρ , both under the COVERAGE-CALIBRATED critical value of Section 6.3 — the nominal ones are in Table S6. THE REGION IS THE LABEL Definition 3 YIELDS, minimum resolved share included: ‘resolved share’ is the percentage of the inference family the band resolves, and a direction is withheld below 5% of it, which withdraws the direction on 4 of the 13 pairs that carry one (Table 7 prints the counts underneath). Last, the share of admissible specifications whose sign disagrees with a conventional one-number report, under the equal-level measure, which is the ‘all-cells rate’ column of Table S7. The reference cell’s sign and the region need not agree, and on several pairs they do not: a surface can be conditionally beneficial while the one cell an analyst would have stood on is negative.

6.1. The master table

Table 5 is the frozen master table: one row per log–target pair, the increment at the reference specification with its simultaneous band, the region label, the robustness index and the sign-disagreement rate of a conventional one-number report. The reference specification is declared and is the same everywhere — one-hot logistic regression, the single temporal holdout, the clean register, the intake block plus the free field as the baseline, ROC AUC — and its region and ρ columns are the coverage-calibrated ones of Section 6.3 while its sign-disagreement column is the all-cells rate of Table S7 under the equal-level measure. **The reference cell’s sign and the region label need not agree, and on several pairs they do not:** a surface can be conditionally beneficial while the one cell an analyst would have stood on is negative, because the label ranges over the whole inference family and the reference cell is one member of it. Table 7 prints the counts each label rests

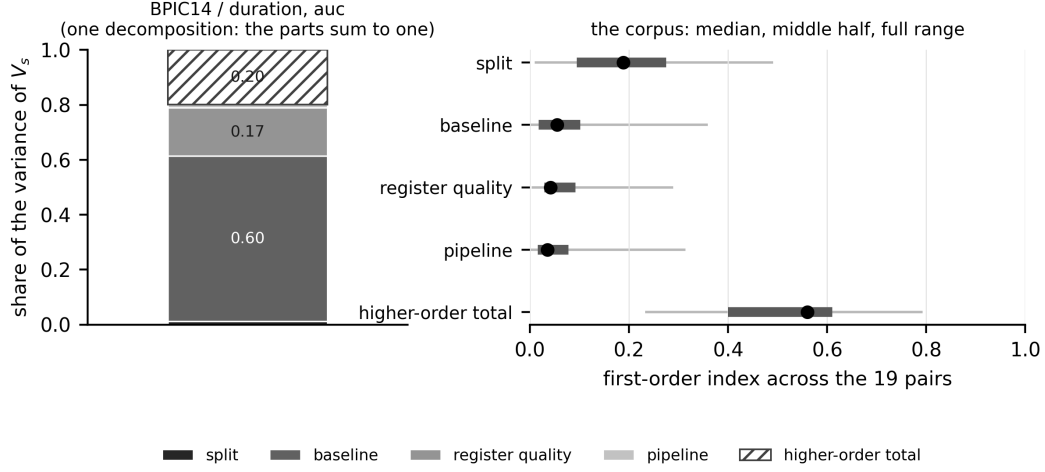


Figure 3: The decomposition, within instrument, under the equal-level measure. *Left*: one pair at one instrument, so the components are a decomposition and the parts sum to one exactly — the largest pair in the corpus, at the reference instrument, which is the cell every other table reports at. The hatched segment is the total higher-order share $1 - \sum_i S_i$ of (4). *Right*: each axis’s first-order index across the corpus, as the median over pairs with the range the middle half occupies and the full range. Nothing on the right is stacked, because a median over pairs and a median over instruments do not add and a stacked bar of them would invite the addition. The bootstrapped indices and their intervals are in Table S5.

on, and Table 4 which attribute plays the register on each row.

6.2. What the choices are worth

Table S5 and Figure 3 give the decomposition, computed within each instrument in that instrument’s own units, under the equal-level measure. The table’s first four columns are, per pair, the median over the five instruments of each axis’s first-order index; its last column is the largest per-axis interaction *involvement*. Three aggregations of this table are quoted below and they are not interchangeable, so each is named where it appears: the median over pairs of a column of this table; the 28.7% figure, which is the median over pairs of the median over instruments of the *per-instrument* largest first-order index, and which is therefore not obtainable by taking a maximum across the four printed columns; and the corpus medians of the higher-order share. The per-instrument quantities the second is built from are in Table S23.

Most of what the pooled surface varies over is not specification at all. Table 6 partitions the decomposition by (3), exactly and on every

log	target	analyst choice	counterfactual	mixed	resampling
BPIC13_closed	handover	31.5	2.9	5.8	59.8
BPIC13_incidents	duration	13.3	2.9	3.0	79.6
BPIC13_incidents	handover	19.2	9.9	4.5	42.3
BPIC14	duration	71.9	15.1	7.9	5.1
BPIC14	handover	73.9	12.7	10.3	3.6
BPIC15_1	duration	9.4	3.9	4.7	80.5
BPIC15_1	handover	13.4	3.6	3.6	62.6
BPIC15_3	duration	11.7	8.1	1.5	77.6
BPIC15_3	handover	22.3	0.4	2.5	76.1
BPIC15_4	duration	11.3	2.5	5.6	81.6
BPIC15_4	handover	12.4	3.1	5.7	71.7
BPIC15_5	duration	17.4	4.2	3.9	69.9
BPIC17	duration	4.2	6.2	5.2	84.1
BPIC19	duration	3.8	1.1	2.8	90.9
Helpdesk	duration	31.7	8.6	1.7	53.6
RoadFines	duration	7.0	28.9	2.1	61.7
Sepsis	duration	13.2	14.5	5.0	71.9
UCI498	duration	11.0	3.0	4.3	81.1
UCI498	handover	41.0	7.1	9.3	42.4

Table 6: THE DECOMPOSITION WITH THE SPLIT AS AN ERROR STRATUM. Per cent of the pooled variance of V_s , per pair, median over the 5 instruments, under the equal-level measure. The four columns are an exact partition of the orthogonal decomposition’s components, and on *each* of the 304 decompositions they sum to 100 to within 4.4×10^{-16} . **The printed rows are medians over instruments and therefore need not sum to 100**, for the same reason Figure 3’s right panel is not stacked: a median is not linear. ‘analyst choice’ collects every component built only from the pipeline and the baseline rung, ‘counterfactual’ the register-quality main effect, ‘mixed’ the components that join the two, and ‘resampling’ every component involving the split — five of whose six levels are expanding-origin folds on the same data. Dropping the sixth, the single temporal holdout, so that the axis is folds and nothing else, moves the last column’s corpus median from 71.7% to 71.0%.

decomposition separately. Taking each column’s median across the corpus — *four medians, which do not add, and are not four parts of one pair* — the components involving the split carry 71.7% of the variance, against 13.3% for the choices an analyst makes, 4.2% for the register-quality counterfactual and 4.5% for the components that join them. Resampling is the largest of the four on 17 of the 19 pairs. Dropping the one split level that is a genuine analyst choice — the single temporal holdout, leaving five folds of one rule —

moves that share only from 71.7% to 71.0%, so the stratum is not an artefact of mixing a rule with its folds. **The two exceptions are the case study’s own pairs**, where the analyst axes carry the richest grids in the corpus — 4 pipelines and 10 register-quality conditions against two and six elsewhere (Table 4) — and the ordering reverses.

Net of resampling, no single axis dominates — and this is a weaker statement than the pooled surface makes it look. On the fold-averaged surface, which carries 28.3% of the pooled variance, the median first-order index is 21.9% for the baseline rung, 19.2% for the register-quality condition and 15.2% for the pipeline, and the largest of the three at the median pair is 38.8%. **The total higher-order share is 26.3% once resampling is removed, against 56.0% on the pooled surface, and it no longer exceeds the largest first-order index:** pooled, the interactions carry more than any main effect on 13 of 19 pairs; net of resampling, on 7. *Most of the variance belongs to no single axis* is a property of a surface that counts five folds as five specifications, and it does not survive treating them as what they are. What does survive is the weaker and still sufficient claim: which axis leads is a property of the pair — the rung on 8, the register-quality condition on 5 and the pipeline on 6 — so an analyst cannot learn from somebody else’s paper which of their own choices their answer turns on.

The pooled decomposition, for comparison. Across the corpus the median pooled first-order index is 18.9% for the split, 5.5% for the baseline rung, 4.2% for the register-quality condition and 3.6% for the pipeline; at the median pair the largest of the four is 28.7%, the total higher-order share is 56.0% (95% interval 40.9% to 60.8% over pairs, 23.3% to 79.3%), and the largest per-axis *involvement* $\max_i(S_{T_i} - S_i)$, a different quantity, is 43.7%. It is reported because it is what the specification-curve literature computes and because the difference between the two is this subsection’s result, not because it answers the question as asked.

The measure is not neutral, and one conclusion does not survive it. Across the three product measures the higher-order share moves by 24.3%, and under the measure concentrated on the reference cell it is 31.7% — smaller than the largest first-order index under the same measure, 39.2%, on 12 of 19 pairs. *The interactions carry more than any main effect* is therefore a claim about a measure and not only about a surface. Supplement S10.10 carries the target axis, which axis leads on which pair, and both invariance tests.

The baseline axis moves the answer by more than the answer, on the pairs where an analyst has more than one baseline to choose between. Holding the pipeline, the split, the register and the metric at the reference and varying only the baseline rung, the increment moves by 0.073 AUC at the median pair and 0.252 at the widest — larger than the increment itself at the reference cell on 15 of the 19 pairs. **That range is carried by the half-of-intake rung, and a reader is owed the spread without it.** Half of an intake block, taken in a deterministic order by cardinality, is admissible but impoverished in the same way the intercept-only rung this paper already excludes is impoverished. Excluding it too, the spread is 0.0055 AUC at the median pair and 0.2495 at the widest, and exceeds the reference increment on 7 pairs. The fall is not a correction of an error but a change of denominator with a plain explanation: on 17 of the 19 pairs only two rungs remain, so the spread over them *is* the absorption D of the free field, which Section 7.5 reports as an object in its own right with its own interval. Only the case study’s log carries a third (Table 4), and its two pairs are where the corpus’s widest realistic-rung spread sits, reaching 0.2495. The index and the spread answer different questions — what share of the *variance* an axis explains while every other axis is also varying, against how far that one axis moves the answer on its own — and both are reported because either alone misleads.

The pipeline axis is two axes, and it does not run at full width everywhere. Its levels are a model family together with an encoding and, in one case, a calibration, and on 15 of the 19 pairs only two of them run. **The admission rule is a function of the log’s row count and nothing else**, declared before any fit: the case study’s log carries all 4, the two largest logs carry the two that differ in *encoding* — a boosting fit on a quarter of a million rows costs more than the rest of the corpus together — and every other pair carries the two that differ in *family*. Table 4 gives the count per pair. Crossed on the case study’s log — both families against all three encodings on both targets, 12 complete factorial pipelines — **the encoding is the larger of the two**, 10.1% of the variance against the family’s 5.2%, with an interaction, 9.7%, as large as either main effect: the fused index is not the sum of two indices. Supplement S10 carries the crossed decomposition. It has not been re-run on the other 15 pairs, which Section 11 records.

6.3. Resolution regions, and the coverage they have

Table S6 and Figure S4 carry the region labels and the robustness index for every pair, computed on the inference surface under the whole-surface simultaneous band at the conservative end of the critical value’s Monte Carlo interval. That family has a median 180 members per pair and a median critical value of 3.33, against 2.35 for a within-instrument family and 1.96 pointwise; under the narrower family the same draws give a different region label on 9 of the 19 pairs, and the whole-surface family resolves 1,150 cells against the narrow family’s 1,510.

The nominal critical value is not enough, and the correction is measured rather than assumed. Section 10.3 shows that coverage is governed by K/n and that this corpus lives where coverage is below nominal, the median pair sitting at 0.106, so a region label computed from a nominal band overstates how much of a surface is resolved. The (n, K) plane is dense enough to calibrate against, and Supplement S9 derives the inflation factor $c(K/n)$ from it — a monotone regression through the measured factors, whose log-linear summary has slope 0.102 (standard error 0.005). The factors the corpus draws range from 1.18 to 1.77; at the median pair the critical value goes from 3.33 to 4.54; the corpus resolves 896 cells against 1,150, a loss of 22.1%, and 4 of the 19 region labels change. **Every region label and every ρ quoted in this paper is the calibrated one;** Table S6 prints the nominal columns beside them.

The factor’s reach, and what widening the plane in n found. It is fitted at training sizes 500 to 180,000 against a corpus running 735 to 176,213, so all 19 of the 19 pairs lie inside the range that identifies it. Two things took the place of the extrapolation that used to sit here. **K/n is not sufficient:** $\log n$ enters the fit at $t = 8.3$, and the applied curve, a function of the ratio alone, sits below the measured factor’s own lower confidence limit on 2 of the plane’s cells. **And on 4 of 33 cells the factor does not exist**, because past a crossing in n the basic interval stops containing the point estimate a widening would be about (Section 4.1). Supplement S9 carries the ladder, the two-covariate comparison and the admission rule; Section 11 carries what it leaves the corpus’s largest pairs.

Of the 13 pairs that resolve anything, 9 have no harmful cell and 3 resolve cells of both signs. That is the informative reading of the region column, and it is the one to carry away: on two thirds of the pairs where the data determine any sign at all, every sign they determine points the same way; on the remaining third they do not.

log	target	family size	beneficial/harmful/unresolved	resolved share	region	label under minimum share
BPIC13_closed	handover	180	0/0/180	0.0	unresolved	–
BPIC13_incidents	duration	180	3/0/177	1.7	conditionally beneficial	unresolved (below minimum share)
BPIC13_incidents	handover	180	76/0/104	42.2	conditionally beneficial	–
BPIC14	duration	480	332/0/148	69.2	conditionally beneficial	–
BPIC14	handover	480	267/2/211	56.0	sign-changing	–
BPIC15_1	duration	180	2/0/178	1.1	conditionally beneficial	unresolved (below minimum share)
BPIC15_1	handover	180	0/0/180	0.0	unresolved	–
BPIC15_3	duration	180	10/0/170	5.6	conditionally beneficial	–
BPIC15_3	handover	180	0/0/180	0.0	unresolved	–
BPIC15_4	duration	180	0/0/180	0.0	unresolved	–
BPIC15_4	handover	180	0/0/180	0.0	unresolved	–
BPIC15_5	duration	180	0/0/180	0.0	unresolved	–
BPIC17	duration	180	25/0/155	13.9	conditionally beneficial	–
BPIC19	duration	120	14/24/82	31.7	sign-changing	–
Helpdesk	duration	180	0/2/178	1.1	conditionally harmful	unresolved (below minimum share)
RoadFines	duration	120	45/18/57	52.5	sign-changing	–
Sepsis	duration	180	51/0/129	28.3	conditionally beneficial	–
UCI498	duration	180	17/0/163	9.4	conditionally beneficial	–
UCI498	handover	180	8/0/172	4.4	conditionally beneficial	unresolved (below minimum share)

Table 7: WHAT A REGION LABEL RESTS ON. $|\mathcal{F}|$ is the inference family’s size; the triple counts its cells the coverage-calibrated whole-surface band calls beneficial, harmful and unresolved. A directional label follows from as few as two resolved cells, and the last column applies the minimum resolved share of Definition 3 — 5% of the family — which withdraws the direction on 4 of the 13 pairs that carry one. A dash means the label is unchanged.

No surface in this corpus is uniformly beneficial, and that statement is weaker than it looks — the label requires every admissible cell to resolve beneficial, and a cell with a negative point estimate can never do so at any critical value (Supplement S10.17). The counts are reported at four calibration settings and the ordering is stable across all of them (Supplement S9.7), so no conclusion here rests on the calibration being exactly right.

6.4. What one number costs

A conventional report — one pipeline, the clean register, the single hold-out, ROC AUC — disagrees in *sign* with a median 32.8% of the other admissible specifications under the equal-level measure. **Three things are wrong with reading that as *what a one-number report exposes a reader to***, and this section corrects each of them, then all three at once: the axes are not all the same kind of thing, most cells are not cells whose sign the data determine, and a disagreement counted at any magnitude is not a disagreement worth acting on. Correcting them one at a time is not enough — each correction moves the denominator the next one is computed on — so the rate that survives is the one taken under all three together.

The axes are not all the same kind of thing. The design space mixes choices an analyst makes with things that are not choices at all, and

Table S17 partitions the decomposition accordingly: *analyst latitude* is the pipeline and the baseline rung; *resampling* is the split, five of whose six levels are expanding-origin folds on the same data; *counterfactual* is the register-quality condition, a different state of the world rather than a different way of analysing this one. On the primary scale, analyst latitude carries a median 11.1% of the variance, resampling 18.9% and the counterfactual 4.2%, and resampling is the larger of the first two on 10 of the 19 pairs.

Whether the split moves the increment more than the analytic axes depends on which scale the partition is taken under, and we report both rather than the one that reads better. The primary scale decomposes *inside* each instrument and takes the median across the five, so the instrument is a stratum and its own contribution is in none of the three columns. Making it a sixth axis needs the headroom rescaling of Section 4.5, because raw AUC and raw average precision do not share units — and on that scale the ordering reverses: analyst latitude carries 19.2%, resampling 9.4% and the counterfactual 2.7%, with resampling the larger on 7 pairs. The instrument is an analyst’s choice, so the second number is the one that answers the question as asked; the first is the one taken on the units the rest of Section 6 uses. Neither is a corpus fact independent of the scale, which is the paper’s own thesis applied to the paper. Restricted to the 30 analyst-latitude cells a pair carries — two pipelines by three admissible rungs by 5 instruments, with the register clean and the split at the single holdout — the disagreement rate is 20.0% against the pooled 32.8%: the first is what a one-number report exposes a reader to, the second what a claim ranging also over degraded registers and resampled splits is exposed to.

The reference specification is itself a choice, and moving it moves the headline by less than the corrections above. The rate counts cells whose sign disagrees with the reference cell, so the reference supplies the sign and nothing else. Varying it one axis at a time over every level that axis carries — 26 variants in all, Table S18 — the corpus median runs 32.8% to 38.4% against the declared 32.8%: 35.3% if the reference pipeline becomes target-encoded boosting, and 32.8% to 38.2% across the five instruments. The headline is not an artefact of which cell we called conventional.

And most cells are not cells whose sign the data determine. The median pair resolves 8 cells of the 180 it carries, and sign agreement between two draws from a distribution centred near zero is one half by construction, so a rate computed over unresolved cells is partly measuring that.

And a sign flip between two increments that both round to zero

log	target	V at reference	above MPID	ref. resolved	all cells	above MPID rate	resolved	resolved and above MPID	mean deviation, AUC
BPIC13_closed	handover	-0.0091	–	–	39.4	–	–	–	–
BPIC13_incidents	duration	0.0278	yes	–	21.8	16.2	0.0	0.0	0.0601
BPIC13_incidents	handover	0.0373	yes	yes	8.3	5.5	0.0	0.0	0.0586
BPIC14	duration	0.0959	yes	yes	8.8	0.9	0.0	0.0	0.0209
BPIC14	handover	0.1673	yes	yes	23.3	21.4	0.0	0.0	0.0689
BPIC15_1	duration	0.0011	–	–	38.4	–	–	–	–
BPIC15_1	handover	0.0071	–	–	22.7	–	–	–	–
BPIC15_3	duration	0.0354	yes	–	7.4	4.8	0.0	0.0	0.1127
BPIC15_3	handover	-0.0030	–	–	74.1	–	–	–	–
BPIC15_4	duration	0.0348	yes	–	56.5	56.3	–	–	–
BPIC15_4	handover	0.0015	–	–	70.8	–	–	–	–
BPIC15_5	duration	0.0320	yes	–	21.8	17.6	–	–	–
BPIC17	duration	0.0021	–	–	23.1	–	0.0	–	0.0215
BPIC19	duration	0.0244	yes	–	17.6	16.4	20.0	20.0	0.0362
Helpdesk	duration	-0.0195	yes	–	37.5	32.4	–	–	–
RoadFines	duration	0.0306	yes	yes	4.0	2.9	0.0	0.0	0.0126
Sepsis	duration	0.1527	yes	yes	0.0	0.0	0.0	0.0	0.0422
UCI498	duration	-0.0069	–	–	61.1	–	100.0	–	0.0438
UCI498	handover	-0.0033	–	–	54.2	–	–	–	–

Table 8: SIGN DISAGREEMENT ONCE MAGNITUDE IS REQUIRED. On the AUC sub-surface, because the minimal practically important difference is declared in AUC and Remark 1 forbids carrying it to another instrument. The four rate columns are per cent of cells whose sign disagrees with the reference cell: over every admissible AUC cell; over those where the reference increment and the cell’s own increment both exceed the MPID of 0.01 AUC; over those the coverage-calibrated whole-surface band resolves; and over cells that satisfy both restrictions. A dash is not a zero — it is a restriction that leaves no cell to compute a rate on. The last column is a magnitude and not a rate: the mean distance from the reference increment over the cells the band resolves, in AUC.

is not a misstatement anybody would act on. The rate counts a cell whenever its sign differs from the reference cell’s, at any magnitude, and 8 of the 19 reference increments are themselves inside a hundredth of an AUC point. A *minimal practically important difference* is therefore declared — MPID = 0.01 AUC — and a disagreement is counted only when the reference increment and the cell’s own increment both exceed it. The MPID is declared in AUC and Remark 1 forbids carrying it to another instrument, so the restriction is applied on the AUC sub-surface and the unrestricted AUC rate is reported beside it, in Table 8.

On the cells the corpus can resolve, the rate is 7.5%, and it is not an AUC statement. Of the 896 cells whose coverage-calibrated whole-surface band excludes zero, 67 disagree in sign with the reference cell. Split by instrument that rate runs 2.5% on ROC AUC to 11.2% on Nagelkerke R^2 : the pooled figure is carried by the instruments that are not the one the reference cell is read in, which is Remark 1 showing up in the corpus.

Imposing every correction this section’s own argument asks for leaves 42 cells, and 1 of them disagrees. Restrict to the axes an analyst chooses, to the cells the band resolves, and to increments that clear the MPID at both ends, all on the AUC sub-surface: 42 cells remain of 5,808, and the

rate is 2.4%. **We therefore withdraw the claim that a one-number report misstates a sign at a rate worth quoting as a headline.** On the corpus’s own AUC surface, once a sign has to be determined *and* worth acting on, disagreement is rare and the denominator is too small to carry a rate. Two corollaries follow and neither is comfortable for the conventional report. First, the reason so few cells remain is that *the data resolve so little*: 197 of 5,808 AUC cells, and 14 of the 19 pairs do not even resolve the cell a conventional paper would stand on. Second, **the surface still moves, and by a distance that dwarfs the MPID**: over the cells the band resolves, the mean distance from the reference increment is a median 0.0430 AUC per pair, and 0.0534 under the full restriction, both several multiples of the MPID. The one-number report is mostly not wrong about the *sign*; it is wrong about the *size*, and it does not say which.

6.5. *Against specification-curve analysis as practised*

The closest existing practice is specification-curve analysis [35]: enumerate the reasonable specifications, plot the curve, and test it against a sharp null by permutation. Table S19 runs it unchanged on *every* admitted pair — 19 of them, 100 permutations of a declared 24-cell sub-surface — beside the region label computed on the same cells, which makes “how often do the two verdicts part company” a measurement rather than an illustration. **That budget floors the permutation p -value at 0.010**, which is ample for the comparison made here — the test either rejects decisively or does not, and no conclusion turns on a p -value near the floor — but a reader wanting to quote a permutation p -value itself needs more permutations than this. Two budgets are declared, the sub-surface and a cap of 20,000 cases per pair that binds on 7 of them; Supplement S10 gives both.

They part company on 9 of the 19 pairs, close to half. The permutation test rejects the sharp null — the surface is decisively unlike one with no signal — and the same cells contain increments of *both* signs. A reader given only the test would conclude that the register matters and have no way to learn that whether it helps or hurts depends on choices nobody told them about. On 3 pairs neither resolves and the cheaper instrument suffices; the rest are mixed.

And a specification curve is a statement about the cells it enumerates. On 1 of the 19 pairs every cell of the sub-surface has one sign while the *full* surface is sign-changing or conditionally harmful. Nothing is wrong with either — they are claims about different sets — which is the

point: without a declared denominator and a simultaneous band, a curve cannot distinguish “the sign holds” from “the sign holds on the cells I drew”.

Neither subsumes the other. The permutation test answers *is there anything here*, costs one refit per permutation per cell and needs no band; the region answers *may I quote a sign*, costs a nested bootstrap, and is what a reader deciding whether to act needs. This paper’s contribution is the second, and the first should be reported beside it.

6.6. Decision rules, in sample and out

The four rules of Section 4.7 are compared under the loss defined there, inside one instrument at a time. Out of sample — choosing on a random half of the admissible cells and paying on the other half — the one-number rule costs 0.00450 AUC of expected excess regret against 0.00011 for the majority rule and 0.00002 for the weighted mean. **Three results run against the surface’s advocate and we report them:** the conservative rule is worst in 4 instruments of 5, the one-number rule’s median excess regret is *zero* in every instrument, and under rolling-origin folds the ordering does not hold, at 0.00885 against 0.01922. We therefore do not conclude that a surface report is a better decision rule than a number; the case rests on the decomposition and the sign disagreement, and this is a check on those. Table S24 and Supplement S10.8 carry it.

7. Case study: a configuration management database

A configuration management database is the register an IT estate keeps of the items it contains, and it is bought partly on the argument that operational analytics need it. This section evaluates one against a named decision on a public log: will this case be reassigned to another group before it is resolved? The estate is a bank’s and the log is BPI Challenge 2014 [12]; the register field is the affected configuration item, with 3,019 distinct values. It is reported at greater length than the other 19 pairs because it is the case that motivated the study and the one on which the decision time — when in the service-desk process the prediction is made — decides the answer.

7.1. One log, two published answers, and what separates them

This log appears twice in this paper. Section 6 applies the registered generic rules of Section 5, which admit 46,606 cases and define the target by

the registered handover rule. This section uses the cohort and the reassignment target defined for the estate: 45,455 cases at prevalence 0.400 (0.372 in the test half), 31,818 in training and 13,637 in test. The first set is a strict subset of the second; the extra 1,151 cases are those the estate’s own cutoff excludes.

The two produce different answers about the register at the later decision time, and the difference is large enough to change the sign. Against intake plus group plus the knowledge reference, the estate’s cohort and reassignment target give -0.0029 and the registered cohort and handover target give $+0.00768$, on which the planned contrast PC3’s one-sided p -value rejects at the Holm-adjusted level while its interval does not exclude zero (Section 4.4). Table S8 takes the manuscript from the first to the second one factor at a time.

It is the target, not the cohort. Over the cohort $\times$ target design the target carries 99.3% of the variance in the increment, the cohort 0.0%, and their interaction 0.7%. Reassignment to another group before resolution has prevalence 0.400; handover between activity groups, under the registered rule, has prevalence 0.927. They are not the same question and the register is not worth the same to them. Two further candidate explanations were checked and neither is one: the two analyses reach the same attribute values on every row, so it is not a data-preparation difference; and they run the same pipeline at the same reference cell, so it is not a modelling difference.

An axis nobody declares, and this paper was standing on it. Both analyses sort the log in time and split at seventy per cent of the row order, and 270 incidents share their timestamp with another; the sort the case study originally used was not stable, so its tie order was neither declared nor reproducible, and it printed -0.0032 where the two declared rules both give -0.0029 . Over 25 random tie-breaks the increment ranges over 0.00088 AUC without changing sign (0 of 25 draws positive), so no conclusion turns on it — but it is a quarter of the point estimate’s own magnitude, produced by a choice no reporting standard asks anyone to state, which is why Section 9 now asks. Every number in this section is computed under a declared tie-break, by incident identifier, on the estate’s cohort and reassignment target; the registered cohort’s answers are in Section 6 with the other 19 pairs, so the decision-time result below is a statement about a routing decision on this estate and not about handover in general.

decision time	baseline	n	prevalence	register levels	base auc	with auc	V	interval	resolved
T0 first touch, every call	intake	145478	0.146	4034	0.554	0.785	0.231	[+0.220, +0.262]	True
T1 first touch, escalating only	intake	44513	0.402	2445	0.552	0.754	0.201	[+0.193, +0.222]	True
T2 incident creation	intake	45455	0.400	2929	0.562	0.746	0.184	[+0.172, +0.204]	True
T2 incident creation	intake + group	45455	0.400	2929	0.644	0.748	0.103	[+0.095, +0.130]	True
T2 incident creation	intake + group + knowledge	45455	0.400	2929	0.805	0.802	-0.003	[-0.007, +0.002]	False
T2 incident creation, T1's population	intake	44513	0.402	2887	0.559	0.750	0.191	[+0.183, +0.213]	True

Table 9: The decision time implemented as an axis. Each moment carries its own population, its own register and its own admissible set; the PREDICTED TARGET is the same at all of them. τ_0 and τ_1 differ only in which cases are in scope. τ_1 and τ_2 differ in what is known AND in which cases are in scope, because 942 incidents have no interaction record; the sixth row is τ_2 restricted to τ_1 's own population, and it is that row, not the third, that makes the τ_1 -to- τ_2 step an information step alone. Every row is on the estate's own cohort and the reassignment target, under a declared tie-break, and every interval is the pointwise basic construction at 300 draws.

7.2. Three decision times

A decision time fixes three things at once — which cases are in scope, what is known about them, and which fields are admissible — and the estate's surface is estimated at each of three moments.

The register's worth turns on which one the analyst occupies: at τ_0 it is +0.2310 [+0.2204, +0.2618] over 145,478 *interactions* — calls to the desk, of which only a minority ever become incidents, which is why that population is far larger than the 45,455 incidents the later moments carry — at τ_1 +0.2013 [+0.1930, +0.2224], and at τ_2 against the intake block plus the group it is +0.1034 [+0.0949, +0.1295] while against the block, the group and the knowledge reference it is -0.0029 [-0.0074, +0.0019] — not resolvably different from zero. **The step from τ_1 to τ_2 is an information step only on the matched population:** 942 of the 45,455 incidents have no interaction record, so the sixth row of Table 9 restricts τ_2 to τ_1 's own cases and it is that row, not the third, that isolates the information. Supplement S10.14 gives each moment's population, register and admissible set.

7.3. The leakage tipping point, and register quality

Two further experiments on this estate are in Supplement S10.4 and summarised here.

A field that looks like leakage has a tipping point, and where it sits is what a reader has to judge. The knowledge reference is the field that takes the register's increment to indistinguishable from zero, and whether it was available at τ_2 is probable but unproven. Writing λ for the share of its values assumed written *after* the incident — so that a larger

λ makes more of the field inadmissible — the register is worth -0.0029 $[-0.0074, +0.0019]$ at $\lambda = 0$, not resolvably different from zero, and reaches half of its knowledge-free value at $\lambda = 0.64$. A reader who believes more than 64% of references are written after the incident should read this estate’s register as worth at least half of $+0.1034$ AUC, and one who believes almost none are should read it as worth almost nothing. We cannot measure λ , so the disposition of an unprovable admissibility assumption is reported as a sensitivity analysis and not as a headline (Figure S2, Table S21).

Register quality is an axis, not a caveat, and which rows go missing matters as much as how many. Under six declared degradation mechanisms the increment at the reference cell moves from $+0.1034$ on a clean register to $+0.0819$ when half the population is lost from its long tail and $+0.0708$ when half is lost from its core. Three of the mechanisms are *dependent* — on time, on content, and on a training-estimated propensity that correlates missingness with the outcome — and they carry severity integrals $+0.0768$, $+0.0486$ and $+0.0559$ against $+0.0692$ for the independent long-tail mechanism, which they bracket rather than track. A quality axis built only from independent coin flips measures one corner of the question.

Which halves are degraded, exactly. A degraded register is a state of the world and not a training accident, so **the mechanism is applied to both halves of the data**: the same values are missing, stale, corrupted or split at fit time and at evaluation time, and the increment measures a model built on a degraded register and deployed against one. What is estimated from the *training half alone* is the mechanism’s *parameters* — which values fall in the long tail, which are common, the frequency table a corruption draws from, the set of identities a reconciliation failure splits — so that no test outcome and no test-half content can reach the counterfactual. That property is executed rather than asserted: the test half is perturbed, the mechanism is re-applied, and the training half’s degraded values are required to be identical, on every mechanism and level. The distinction matters because the two designs measure different things, and the earlier wording of this paragraph did not make clear which was in force: a mechanism applied to training alone would measure a train–test mismatch, which is not the question a register’s owner is asking. Tables S28 and S29 carry every mechanism and level.

7.4. What this says to a configuration management programme

This subsection is the reading a service-management audience will want, stated in the estate’s own vocabulary. It is a reading of *one* estate on *one*

public log and it is offered as one; Section 11 says what a second would need.

Which field is which. The prediction is: *will this case be reassigned to another group before it is resolved?* The **register** f is the affected configuration item, 3,019 distinct values. Its **coarsenings** are the item’s type (13 values) and subtype (65), each given its own role rather than swept into the baseline. The **free field** g is the assignment group — the routing queue, which is recorded without a configuration management database and is the field the register has to beat. The **intake block** B_0 is what the desk records when the case arrives: category, impact, urgency and priority. The **knowledge reference** is a pointer to a knowledge-base article, admitted as a further rung because whether it is available at the moment of decision is exactly the question Section 7.3 cannot settle.

The three moments, in service-desk terms. τ_0 is *an interaction is logged*: somebody has contacted the desk, every call is in scope, and 145,478 of them are. τ_1 is the same moment restricted to the calls that will escalate — the same information, a different population. τ_2 is *an incident record is created*: 45,455 cases, more known about each, and 942 of them with no interaction record at all. Table 9 is the ladder. **Read down it and the register’s worth falls from +0.2310 AUC to +0.2013 to +0.1034 once the routing queue is in the baseline, and to −0.0029 — not resolvably different from zero — once the knowledge reference is admitted too.** A programme that justifies the register on a first-touch number and then deploys the model at incident creation is quoting the top of a ladder and standing on the bottom of it; Section 8.4 prices that in cases per thousand: 1.71 true positives promised per thousand cases against 0.003 delivered.

Which layer of the register pays. A programme does not choose whether to have a configuration management database; it chooses how much of one to populate and maintain, and that is a choice among *layers*. **The ladder is not Table 9’s cell and does not reconcile with it arithmetically:** it is computed on the *registered* cohort of Section 5 — 46,606 cases, 13,982 in test — under the registered handover rule, one-hot logistic regression at the fixed temporal holdout, against the intake block at base AUC 0.641, whereas Table 9 is on the estate’s own cohort and its reassignment target. Section 7.1 is the reason the two differ: those are the two analyses whose answers about this register disagree, so the register’s increment over intake is +0.257 here and +0.1838 there, and neither number is the other one mismeasured. What

the ladder compares is layers *against each other* at one cell, which is the comparison a programme is choosing among. On it the item's *type* is worth +0.134 AUC over intake, its *subtype* +0.163, and the item itself +0.257; with the subtype already in the model the item still adds +0.109. **On this estate and this question the coarse layers are not free, and the finest layer is not redundant:** about half of what the item is worth over the intake block is already reachable from a thirteen-value type field, and the item still adds +0.109 once the subtype is in the model. On the duration question — the registered duration rule on the same cohort, base AUC 0.618 — the same ladder gives +0.045, +0.045 and +0.129, with +0.087 marginal — there the coarse layers reach about a *third* of the item's increment and the item carries the rest. **So the split between the layers is a property of the target and not of the register:** the same three fields on the same estate divide their worth one way for reassignment and another for duration. Supplement S4 carries both, and two further logs on which every coarse layer is worth essentially nothing while the finest one is not. In the vocabulary of a layered service model — technical service, application service, configuration item — the finding is that the layers are not substitutes and that which one pays is a property of the question being asked, not of the estate. It is not a procurement conclusion: nothing here prices what a layer costs to populate or to keep accurate, and Section 11 says so.

What would have to change for the answer to move. Three things, in the order a programme could act on them. **A timestamped field history.** The knowledge reference decides this estate's answer and its admissibility is settled here by probability rather than by evidence; a register that records when each field value was written turns Section 7.3's tipping-point analysis into a measurement. **A complete population at the decision time.** 942 of the incidents have no interaction record, and the τ_1 -to- τ_2 step is an information step only on the cases that have both; a desk that can identify its own cases across the two systems recovers that comparison. **A target the desk actually acts on.** The register's sign changes with the target on this log (Section 7.1), and reassignment is the question this desk routes on; a programme evaluating a register against a generic handover rule is answering a question nobody asked it.

7.5. *The absorption, and why the ratio is secondary*

The quantity a report of this kind usually leads with is the *relative* reduction R : the share of the register’s apparent value that a single free field absorbs. The absolute absorption $D = V(f \mid B_0) - V(f \mid B_1)$ is reported first, with its own interval, and R second with a Fieller set and the set’s kind. On this estate at the reference cell D is +0.0804 [+0.0626, +0.0914] and R is 0.437 with a bounded Fieller set. This ladder is the favourable case: 0 of the 30 reductions it produces have a Fieller set that is not a bounded interval, because the denominator is resolvably positive at every rung of it. Across the corpus that is not so — 5,282 of 10,584 reductions have a non-bounded set, and a percentile interval reports a bounded one for every one of them — which is why the ratio is secondary everywhere and not only here. The full ladder is Table S30.

8. Decision-analytic evaluation

An increment in AUC is not a quantity a service desk can act on. This section reads the same surface in net benefit, which is: the number of correctly nominated cases per thousand arrivals, net of the false nominations, at a declared exchange rate between the two.

8.1. *The operating point is a cost ratio, not a knob*

At threshold θ the net benefit of a rule that nominates when $\hat{p} \geq \theta$ is

$$\text{NB}(\theta) = \frac{\text{TP}(\theta)}{n} - \frac{\text{FP}(\theta)}{n} \cdot \frac{\theta}{1 - \theta},$$

so $(1 - \theta)/\theta$ is the number of false nominations the desk is willing to accept for one true one. Choosing θ is choosing an operational cost ratio, and it belongs to the analyst’s declaration rather than to a grid search. We report the whole grid — 31 points from 0.05 to 0.80 — and name three points a desk might occupy: $\theta = 0.10$, or 9.0 false nominations accepted per true one, where a review is cheap and the desk nominates liberally; 0.30, or 2.3, where a review and a miss cost about the same; and 0.60, or 0.7, where review capacity is scarce. A reader whose desk sits at a ratio not in that list reads their own point off Figure S3: the operating point is the reader’s to choose, so the analyst reports the curve rather than a point on it.

8.2. Calibration comes first, or the threshold means nothing

A threshold is a statement about probability, so a model whose probabilities are wrong is being read at a threshold it does not have. Every decision curve here is therefore computed on *calibrated* scores, with the calibrator fitted inside the training half and rebuilt inside every bootstrap refit, and a registered rule excludes an arm whose calibration slope leaves $[0.5, 2]$.

Why that window, and why the exclusion is not repaired by recalibrating again. A Cox calibration slope of γ means a threshold θ selects the cases a well-calibrated model would select at a different threshold, and the discrepancy grows without bound as γ leaves one; the window is the conventional band in which risks are neither compressed nor stretched beyond a factor the literature treats as readable, and it is a declared convention rather than a derived bound. Widening it to admit everything would not make the excluded thresholds readable as cost ratios; it would only stop saying so. **Recalibration is already an axis, which is why it cannot also be the remedy:** isotonic recalibration is a level of the pipeline axis, so an arm that fails the slope check has, on the pairs that carry it, a recalibrated sibling in the same surface. Applying isotonic recalibration to the failing arm before excluding it would silently replace one specification by another and report the result under the first one's name — the exact substitution this paper's argument is against. The arm is excluded and its sibling is reported as itself. Supplement S10.5 gives the rule, what it removes and what the curves look like without it.

8.3. Simultaneous bands change what can be said

The decision-curve family is declared separately from the scalar surface and carries its own max- t critical value, because net benefit and a scalar instrument are not the same scale. Across the 31 operating points of this estate's curve **the pointwise interval declares the increment resolvably positive at 24 of them and the simultaneous band at 18**, and the harmful counts are 0 against 0. Both are built from the same draws, so the only thing that differs is the family: those points are the price of a statement about a *range* of thresholds rather than about each one separately, and a paper that reads a decision curve across its range while pricing it pointwise has claimed the first and paid for the second.

This family is also the one most exposed to how the critical value was estimated. It is *not* coverage-calibrated — the (n, K) calibration of Section 6.3 is estimated for the scalar family — and Section 10.4 finds

that the widening it would need is about twice what the surface families need. Supplement S10.6 carries the comparison, the dip near $\theta = 0.55$ and Figure S3; Table S22 prints the curve.

8.4. *The decision, in units a service desk uses*

The comparison is run in net benefit per thousand cases, over a declared distribution of exchange rates a desk plausibly holds (0.08 to 0.36, centred 0.19), and only on models that pass the registered calibration rule — which excludes 40 of 118 arms and 3 of 19 pairs. **That count is itself the strongest operational finding here:** a decision curve read off a model whose probabilities are wrong is read at a threshold the model does not have.

And the rules mostly tie. The one-number rule’s expected excess is 0.000 per thousand against the conservative rule’s 0.048, the four rules separate on only 11 of 19 pairs, and the one-number rule is worst on 10. **The case against one-number reporting does not rest on this design.** What does separate in this unit is the decision time: the register promises 1.71 per thousand at the earlier moment and delivers 0.003 at the later one, a shortfall of 1.70. Supplement S10.9 carries the rules, the exchange-rate distribution and the exclusions; Tables S20 and S24 the numbers.

9. A reporting standard, and software

9.1. *The minimum reportable form*

The apparatus of Sections 3 and 4 implies a reporting standard short enough to state in full. A feature-value claim is reportable when it carries: **the design space**, as an enumerable declaration of which learners, encodings, targets, splits, decision times, quality mechanisms, baselines, metrics and operating points are admissible, why anything defensible was left out, and *the measure placed on those levels*; **the denominator**, as a count a reader can reproduce from the axis levels — declared cells, computed cells, and the cells any inference actually rests on, which need not be the same set; **the absolute increments** at every rung of the baseline ladder, in the metric’s own units, with intervals from a resampling scheme that refits the model; **the sensitivity decomposition**, with the total higher-order share reported as $1 - \sum_i S_i$ and not as any per-axis difference; **the resolution region and ρ** , under a band whose family is the set the label ranges over;

and **the ratio R last if at all**, with a Fieller set and the set’s kind, never where the denominator is not resolvably positive.

Five further items, each of which this study got wrong before it got right. *The tie-break rule* on the sort that produces the split: an undeclared one moved this study’s own case-study increment by a quarter of its magnitude (Section 7.1). *Which halves a quality counterfactual is applied to*, and which half its parameters are estimated from (Section 7.3). *A minimal practically important difference*, declared before any disagreement rate is computed and in the units of one named instrument (Section 6.4). *Which axes are resampling rather than analyst latitude*, so that a decomposition is not read as specification sensitivity when it is sampling variability (Section 4.5). And *the draw count and the effective draw count* behind every band, together with whatever is known about the coverage those draws buy — because a band reported without them invites a reader to assume its nominal level, which Section 10.4 shows is not attained here. Nothing in the list is expensive once the surface exists, and the surface is a factorial over declarations the analyst has already made implicitly.

Applying it to a field of your own, in six steps. **One**, name the roles: the field f whose worth is in question, the cheap field g it must beat, and the intake block B_0 available at the decision time. **Two**, declare the space: list every level of every axis you would have accepted *before* seeing a result, and the measure over them. **Three**, compute the surface — one call, one cell per specification — and audit the denominator by multiplying the level counts and comparing with the row count. **Four**, decompose it with the resampling axes held as an error stratum, and report the analyst-choice shares as primary. **Five**, band it over the family your claim ranges over, not over one cell, and state the draw count and what coverage those draws attain. **Six**, report the region and ρ with the triple of counts they rest on, and refuse a direction below your declared minimum resolved share. The package below performs steps three to six from a tidy frame.

9.2. *fieldvalue*

The standard ships as `fieldvalue` [11], which computes the surface from a data frame, a feature name and a dictionary of baselines. Its notable property is a refusal: `Surface` raises rather than returning a scalar, `Surface.report()` emits the form above, and `Surface.at(...)` returns one

cell only if the caller names every axis. Version 0.2.0 implements Sections 4.5–4.7 as four functions of a surface — **decompose**, **regions**, **robustness** and, for the robustness check of Section 4.7, **regret** — each taking a tidy frame rather than this package’s own object, so they apply to a surface computed by some other pipeline. It carries 108 tests and is an *independent implementation*: the analysis scripts and the package compute the same quantities from the same definitions in different code, neither written against the other, and on this corpus they agree on 266 of 266 quantities to 1.1×10^{-16} .

9.3. Beyond process logs, and beyond this paper’s claims

The standard is not about event logs and not about configuration databases. Supplement S7 runs it on UCI Adult, where the expensive field is **occupation** and the cheap ones come with the sampling frame: over 480 cells the reduction runs 0.200 to 0.947; the baseline and the metric carry the same share of the variance to three figures, 25.9% and 25.9% — a coincidence rather than a copy error, as the fourth figure shows: 25.94% against 25.90% — against 7.2% for the register population; and a one-number report misstates the sign for a mean 24.9% of the other cells. `examples/worked_example.py` produces all of it in 121 statements against the package’s public interface.

How the surface is reported today. Whether the axes this paper varies are varied in published work deserves a systematic review, and we did not conduct one: a machine-assisted prevalence pilot was run on a convenience frame, is *not* reported here because one adjudicator and a screen of measured sensitivity 0.474 can generate a hypothesis about practice and cannot test one, and is in the archive with its frame, sample, codes and evidence dossiers. The paper’s argument does not need it.

10. Simulation against a known answer

Everything so far has been measured on real logs, where the answer is not known. This section measures the estimator where it is. The full tables are in Supplement S11 and the diagnosis of the one world that failed is in Supplement S6.

10.1. Six worlds, two estimands

The simulation uses six worlds — four of them misspecified for the estimator — at 1,000 replicates each, with the covariate space finite so that the population AUC of any scoring rule is computed in closed form rather than simulated. Two estimands are separated because under misspecification they differ: V_{oracle} , the increment between the two Bayes-optimal rules, and V_{limit} , the increment between the two fitted pipelines’ limits. A resampling interval can only cover the second, and reporting its coverage of the first as the estimator’s coverage is a category error. Supplement S6.1 describes the worlds and the marginalisation a corrupted register forces; Table S34 reports both estimands.

10.2. Coverage, bias and width

Coverage estimates carry a Monte Carlo standard error of about 0.7 percentage points at 1,000 replicates, which is the resolution every comparison here should be read at; Figure S5 draws them by construction and world.

Where the refit helps, and where the percentile interval fails. The nested basic interval covers a median 93.0% against the fixed-model interval’s 92.1%; on the sparse world the *percentile* interval collapses to 37.2% and the basic construction restores it to 89.9%, which is why every interval and band here is the basic one (Supplement S6.2; Table S34).

What no interval repairs. Against V_{oracle} rather than V_{limit} , every construction undercovers on the drifting world — 65.8% at best — and it should: when the coefficients move between eras the estimator is not consistent for the oracle quantity, and the largest gap is 0.0161 in AUC units. Under misspecification an interval around the estimator’s own limit is not an interval around the quantity a reader cares about, and no resampling scheme repairs that.

10.3. Sample size, register cardinality and block length

A construction that works at one sample size, one register cardinality and one block length has been validated at a point, not in a regime. Sample size and register cardinality are therefore varied *jointly*, up to $K = 3,019$ levels — the cardinality of the case study’s own register — and the block length is varied around its rule of thumb.

The ratio K/n captures most of the dependence, and not all of it. Across the plane the basic interval covers a median 83.3% and a minimum 21.3%, the last at a ratio of 0.377, and it is short of nominal everywhere. It is

not sufficient: two cells at nearly the same ratio cover 0.780 and 0.613, a gap of 3.2 combined Monte Carlo standard errors, and holding the register fixed while moving the training half from 2,000 rows to 8,000 moves coverage from 0.453 to 0.613. **This corpus sits in the regime where that matters:** its median pair has $K/n = 0.106$ and 10 of 19 pairs are above a tenth, so a nominal band on them is *narrower* than a 95% interval should be. Section 6.3 calibrates against the plane and Supplement S9 gives it in full.

Where the interval stops containing its own estimate. The denser plane the coverage calibration of Section 6.3 is fitted on — a different and larger object from the 10-cell grid above, at 500 to 180,000 training rows — locates the crossing Section 4.1 describes. At a fixed ratio of 0.125 the basic interval straddles its own point estimate in every replicate up to 8,000 training rows and in only 1.6% of them by 50,000, because the bootstrap displacement is a bias that does not shrink with n while the half-width does. Past that crossing the inflation factor is not defined — it is a widening about the estimate, and there is nothing to widen about — and 4 of the plane’s 33 cells have none for that reason, every one of them at 50,000 rows or more. Section 6.3 reports what that leaves the corpus, and Supplement S9 describes that plane’s design.

Block length. 3 lengths are examined — half the rule $\lceil n^{1/3} \rceil$, the rule, and twice it. Coverage moves with the length, spanning 3.6% within a world, of the order of two standard errors; but the comparison the paper rests on is unchanged at every length, the basic interval beating the percentile interval on the sparse register by at least 46.4% at all 3.

10.4. *The band’s family-wise coverage*

Everything above validates a *pointwise* interval. Two of this paper’s three reporting objects are functions of a *simultaneous* band, and until this section nothing measured what that band covers.

The design, and why its answer is an upper bound. The truth is zero at every cell of a synthetic family. One draw from the sampling distribution gives the point estimate, and as many further draws *from that same distribution* as the corpus family it is matched to carries give the bootstrap — so the bootstrap is not an approximation of the sampling distribution but is it. Every way a resampling scheme can be wrong — dependence it fails to reproduce, a refit that comes apart, a block length misjudged — is switched off, and what is left is the object under test: a standard error and a critical value both estimated from the same handful of draws. A coverage measured here

is therefore the most the real procedure can attain, and a shortfall here is one that no improvement to the resampling repairs. The families are matched to this corpus in size, draw count, cross-cell correlation, effective rank and per-cell excess kurtosis; the matching is fitted by grid search against the corpus's own draw files and reported beside its target rather than asserted. Coverage is over 2,000 replicates in each of 18 cells, so no coverage here is determined worse than 1.1%.

Three regimes, because “the draws are heavy-tailed” turned out to name two different things. Under *Gaussian* draws the multiplier's family-wise coverage is a median 87.3% and a minimum 74.8%. Under draws carrying this corpus's tails, whose median cell has an excess kurtosis of 0.32 on the surface families and whose ninetieth percentile is 3.6, the median is 84.6% and the minimum 76.1%. Under the third regime, which adds the degenerate cells described below, they are 84.4% and 39.2%.

What is wrong with the decision-curve families is not a tail. At a threshold where both arms treat every case alike the net-benefit difference is zero by arithmetic, and 841 cells of the corpus's curve families are exactly zero in at least nine draws of ten, against 0 anywhere in the surface families. They are what makes q_{emp} large there; excluding them moves the median ratio q_{emp}/q from 2.09 to 1.80 and does not dissolve the disagreement. Supplement S9.6 carries the mechanism and Table S37 the regime with and without them.

How far short, in the units the calibration works in. A critical value is widened multiplicatively, so the useful summary of the shortfall is the factor that would have closed it. On the surface families that factor is 1.05 to 1.24; on the decision-curve families, 1.90 to 2.05. The coverage calibration of Section 6.3 applies 1.09 to 1.88, is fitted to restore *pointwise* coverage rather than this one, and is applied to the surface families only — so it is of the right order where it acts and absent where the shortfall is largest. Section 11 says what follows.

The draw count is the binding constraint. At the modal surface family the multiplier band covers 82.0% at 33 draws and 96.6% at 1,000, and the per-cell interval underneath it moves 90.3% to 94.9% over the same range: part of what the band loses it loses before any critical value is chosen, and Section 10.2 validated that construction at a refit count this corpus does not reach on most pairs. No estimator recovers the difference. The best of the five, the empirical quantile at the upper end of its own order-statistic interval, reaches 92.2% at 1.52 times the width and still falls to 86.5% at the

smallest family; a Rademacher wild multiplier, which preserves the observed draws' magnitudes instead of replacing them by Gaussians and is the natural repair for a tail problem, is worse than the Gaussian multiplier at 80.7%.

What the simulation does not establish. It certifies the pointwise constructions on six worlds, a plane of 10 (n, K) cells and 3 block lengths, and the band on the 18 matched families of Section 10.4 — on nothing else. The stationarity and mixing the moving-block bootstrap needs are assumed and not tested; the split point is fixed within a draw, so split-position uncertainty is outside every interval reported here; and the drifting world shows what happens when the estimator is not consistent for the quantity a reader cares about.

11. Threats to validity

Organised as threats rather than as a list. Nothing here is re-derived: every number is stated where it is computed and referenced from here.

Construct — this is predictive performance, not value. Nothing here measures what a register costs to acquire or maintain, whether acting on a prediction changes an outcome, or what a data programme is worth; that includes Section 7.4's layer comparison, which prices no layer's upkeep. The corpus is also a family of related questions rather than replications — the two registered targets label the same case identically on a median 51% of cases, and the registers run from a configuration item to a legal article — so every corpus statistic is about *sensitivity* and every one is reported on the 8 ITSM pairs as well (Table S14); Table S10 names each pair's own construct threat. The registered handover target on the case study's log is close to degenerate at prevalence 0.927, it carries 4,800 of 29,040 cells, and Section 7.1 shows the register's sign turns exactly there. Whether a generic handover rule is a meaningful target on ITSM logs, where reassignment is what a desk acts on, and whether the prevalence window should close below 0.95, are open; we did not narrow the rule retrospectively, because doing so after seeing which pair it admits is what a registered protocol exists to prevent. The quality mechanisms are applied to both halves with their parameters estimated from training alone (Section 7.3), so they are the operational counterfactual and not a train-test mismatch — but they remain *constructed*, and validating them against a genuinely incomplete register is the obvious next study.

Internal — the band does not attain its nominal level. This is the sharpest limitation in the paper. Section 10.4 measures family-wise coverage against a known answer over matched families whose bootstrap is *ideal*, so every coverage there is an upper bound on what the procedure attains: median 84.6% and minimum 76.1% against a nominal 95%, and a median 87.3% even under the Gaussian draws the construction is derived for. The safeguard we called conservative moves coverage by less than a point, because it controls the Monte Carlo error in q and not the error in what q estimates. The repair is more draws — 82.0% at 33 against 95.8% at 400 — and a draw refits the whole design space. **Every region label and every ρ here is therefore a descriptive diagnostic and not a guarantee**, which the front matter now says. The two families are not equally exposed: the widening that would close the gap is 1.05 to 1.24 on the surface families, where the calibration already applies 1.09 to 1.88 — fitted for *pointwise* coverage, so its being of the right order is fortunate rather than by design — and 1.90 to 2.05 on the decision-curve families, **which the calibration does not touch at all**. Those conclusions are the ones to distrust.

Internal — the refitting bootstrap is the source of a bias, and the repair is not run here. A multinomial resample holds about $1 - e^{-1}$ of a high-cardinality register’s levels, so the bootstrap distribution is displaced: the percentile interval becomes a correct interval for the wrong quantity, and the pivotal interval this paper uses can exclude its own point estimate, on 74 of 3,900 cells. The displacement is a bias, so it worsens with n , and it leaves the coverage calibration’s estimator undefined on 4 of 33 plane cells. The design’s own justification is weak: the nested interval beats the fixed-model interval by 1.3 Monte Carlo standard errors, and the seven-construction comparison that chose the pivotal interval ran at 4,000 training rows and at no other size — below the regime where the defect appears. **The repair we would make first is a change of resampling scheme, not of interval**: drawing *weights* — Dirichlet or independent exponentials, applied per moving block so the temporal dependence survives — gives every row positive weight in every draw, so every register level is present in every refit and the displacement disappears at source rather than being pivoted around. That is a change to one function and a re-run of the whole inference surface; the comparison against the multinomial and m -out-of- n resamples across the (n, K) plane is not run, and no conclusion here should be read as evidence that the refit’s cost was worth paying.

Internal — further assumptions, named. The split point is fixed within a draw, so split-position uncertainty is outside every interval here. The moving-block construction needs stationarity and mixing that an event log is not guaranteed to have, and the same assumption is what makes Definition 2’s population limit well defined under a temporal split; neither is tested. The band’s family-wise coverage is asymptotic, as is the multiplier bootstrap estimating its critical value, and the block length is a convention, $\lceil n^{1/3} \rceil$, which moves coverage by up to 3.6% within a world. The coverage calibration is itself an assumption — it corrects the studentised statistic’s *scale* and not the shape of the maximum’s distribution, it is fitted pointwise and applied simultaneously, and it is measurably misspecified ($\log n$ enters at $t = 8.3$) — so a reader who rejects the transfer should rely on the objects that need no band: the point estimates, the decomposition, the disagreement rates and the regret comparison.

Conclusion — two claims a specification surface invites and this evidence does not support. The pooled decomposition is mostly resampling. Net of it the higher-order share falls from 56.0% to 26.3% and no longer exceeds the largest first-order index, so *most of the variance belongs to no single axis* — the reading a pooled surface invites — holds on 7 of 19 pairs rather than 13 (Section 6.2). What survives is that *which* axis leads is a property of the pair. And a sign disagreement is not a misstatement unless the sizes are worth acting on: under the three restrictions the argument implies, the corpus retains 42 cells of which 1 disagrees, so **the sign-disagreement rate is withdrawn as a headline** (Section 6.4). What replaces it is a magnitude and a scarcity: resolved cells sit a median 0.0430 AUC from the reference, and only 197 of 5,808 AUC cells resolve at all, with 14 of 19 pairs failing to resolve even the cell a conventional report stands on. Both decompositions are reported under all three declared measures because there is no neutral measure — across the three the higher-order share moves by 24.3% — and the within-instrument decomposition is primary because the decomposition inherits the metric’s scale (Proposition S1). Region labels remain relative to an inference family of a median 12.5% of a pair’s cells, now under a minimum resolved share that withdraws the direction on 4 of 13 pairs, and they are not partitioned by axis kind, so they stay conditional on the quality axis being averaged over. Specification regret reports a null and is retained as a robustness check rather than a contribution.

External — the design space is declared, not complete. $\mathcal{S}$ has nine axes in Definition 1 and the per-pair factorial varies 5 of them (Table 4); the decision time, the operating point and the cohort vary on the case study’s log alone, and the target indexes surfaces rather than varying within one. It could have more — tuning protocol, missing-data treatment, target horizon, label-noise model — so the honest claim is “the increment moves this much over *this* space”. **The structural omission is that prefix length, prefix bucketing and sequence encoding are not axes here and cannot be**, because this study predicts at one prediction point from creation-time attributes, and those are the axes predictive process monitoring varies most. **We therefore claim no result about that literature’s own benchmark:** what is reported is case-level outcome prediction from event-log attributes, and adding a prefix axis on at least one log is the clearest next study this paper implies.

External — three narrower limits, named. The *inference surface* is a corner of the declared surface chosen by row count rather than a design: axis-complete but unbalanced, the decomposition cannot be computed on it, and its positive-cell share differs from the declared surface’s by a median 0.067; a resolution-IV fractional factorial of the same size would remove both objections and free budget for draws. The *pipeline axis* is crossed on one log only, so the other 15 pairs carry family and encoding confounded. And the *case study* is one bank’s estate on one public log, with the availability question at its centre settled by probability rather than by a timestamped field-value history; Section 7.4 names the three changes that would move which answer. Finally, we do not measure reporting practice: the pilot that attempted it is in the archive, and it is a hypothesis rather than evidence.

12. Conclusion

The incremental predictive performance of a recorded field is a functional of design choices, not a number. This paper made that operational: a declared design space, a surface over it, an exact decomposition of the surface’s variance that separates the choices an analyst makes from the resampling the design also contains, simultaneous bands whose coverage is measured rather than assumed, and region labels that say where a sign is determined and withhold a direction where too little of the surface resolves to support one.

On every pair, at least one admissible specification makes the register worth less than nothing. That is a statement about point estimates and needs no band. It is also nearly guaranteed by a space that

includes three-quarters-deleted registers, so the version worth carrying is the restricted one: **on 7 of 19 pairs the baseline alone — among the baselines an analyst would actually build — moves the increment further than the increment itself.** Four further results are worth carrying away.

Most of a pooled specification surface is resampling, and what remains is not dominated by one choice. The components involving the split carry a median 71.7% of the variance. Net of them the largest first-order index at the median pair is 38.8% and the higher-order share 26.3%, and which axis leads is a property of the pair. An analyst cannot learn from somebody else’s paper which of their own choices their answer turns on — but neither should they read a pooled specification curve as evidence that everything matters.

A one-number report is usually right about the sign and usually silent about the size. Once a sign has to be determined by the data *and* worth acting on at 0.01 AUC, 1 of 42 admissible *AUC* cells contradicts the conventional report — the MPID is declared in AUC and Remark 1 forbids carrying it further. What the conventional report does not say is that the cells it does not stand on lie a median 0.0430 AUC away from the one it does, and that on 14 of 19 pairs the data do not determine the sign of its own cell at all.

The decision time is an argument of the estimand, not a preliminary. It fixes which cases are in scope, what is known about them, and which fields are admissible. On the case study, moving the prediction from first touch to incident creation costs -0.0297 AUC through the population, a further -0.0099 through the information once the population is held fixed, and -0.0076 more through the incidents that have no interaction record at all — before any baseline changes; and the same log’s answer changes sign when the target changes, with 99.3% of the difference on the target and 0.0% on the cohort. A desk that adopted this register on its *first-touch* number and deployed it at incident creation would be promised 1.71 true positives per thousand cases and receive 0.003.

And a threshold is not always a cost ratio. On 40 of 118 fitted models the calibration slope falls outside $[0.5, 2]$, so the threshold at which a decision curve is read cannot be interpreted as the cost ratio the decision-analytic argument requires (Section 8.2).

The practical recommendation is narrow and cheap once the surface exists: report the surface, the decomposition with the resampling stratum sep-

arated, the region and ρ with the triple they rest on; state the decision time and the target as parts of the estimand; declare a minimal practically important difference before reporting any disagreement rate; make the absolute increments primary; and use a ratio last, with a Fieller set, and only where its denominator is resolvably positive.

CRedit authorship contribution statement

Sudhir Dixit: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author has no affiliation with, and received no support from, any vendor of configuration management, IT service management or process mining software.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Anthropic Claude (claude-opus-5, 2026-08-24) in order to draft and edit prose and to write analysis and verification code. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

This is a use beyond the language-and-readability editing an author may make without comment, which is why it is declared. Two further facts belong with it and are recorded rather than summarised: every session's model identifier and date are in `AI-USE.md` in the archive, because a disclosure naming a product and not a version is not reproducible; and **no generative**

model produced, imputed, augmented or selected any datum, any result or any citation. Every result in this paper is produced by public code that `REPRODUCE.md` re-runs by one command, and the reference list was checked against the published records. The one use of the assistant *inside* a study — adjudicating the literature-prevalence pilot, which is in the archive and carries no claim in this paper — is described where that pilot is, in Section 9.3.

Data availability

All data are public. Event logs are obtained by DOI: BPI Challenge 2014 (<https://doi.org/10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35>), BPI Challenge 2013 incidents (<https://doi.org/10.4121/uuid:500573e6-acc4-4b0c-9576-aa5468b10cee>), and the further public logs whose DOIs and SHA-256 checksums are listed in `data/corpus/CHECKSUMS.txt` — 26 files are downloaded and checksummed in all, of which the registered role-assignment rules of Section 5 admit 13, yielding the 19 log–target pairs analysed here. The counts are different quantities and Table S9 accounts for every downloaded log that is not among the admitted pairs, in one of two ways: with the registered exclusion code the rules gave it, or with the code `HELD_OUT`, which marks the held-out set — 1 of the checksummed files, downloaded for the out-of-corpus test and never offered to the admission rules. The literature pilot’s frame, sample, codes, adjudications and evidence dossiers are in the archive; the retrieved full texts are other publishers’ copyrighted PDFs and are not redistributed, but `scripts/s06_audit2.py --fetch` re-retrieves them from the DOIs in `results/s06_sample.csv`. All derived results (`results/*.csv`), all figures and all code are in the archive below.

Code availability

Code, derived results, the pre-registered protocols, the amendment log and the verification harness are archived at the archived release cited in the data-availability statement (DOI reserved, inserted at proof) (release v26.0) and developed at <https://github.com/sudhirdixit1/cmdb-routing-queue-baseline>. The archive contains a `Dockerfile` and a `requirements.lock` pinning every transitive dependency; `REPRODUCE.md` gives the exact commands and the expected runtime.

References

- [1] Amaral, C., Fantinato, M., Peres, S.M., 2018. [dataset] incident management process enriched event log. UCI Machine Learning Repository. doi:10.24432/C57S4H.
- [2] Bouthillier, X., Delaunay, P., Bronzi, M., et al., 2021. Accounting for variance in machine learning benchmarks, in: Proceedings of Machine Learning and Systems (MLSys).
- [3] Breznau, N., Rinke, E.M., Wuttke, A., et al., 2022. Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty. Proceedings of the National Academy of Sciences 119, e2203150119. doi:10.1073/pnas.2203150119.
- [4] Çarka, J., Esposito, M., Falessi, D., 2022. On effort-aware metrics for defect prediction. Empirical Software Engineering 27, 152. doi:10.1007/s10664-022-10186-7.
- [5] Chernozhukov, V., Chetverikov, D., Kato, K., 2013. Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. The Annals of Statistics 41, 2786–2819. doi:10.1214/13-AOS1161.
- [6] Comuzzi, M., Kim, S., Ko, J., Salamov, M., Cappiello, C., Pernici, B., 2025. On the impact of low-quality activity labels in predictive process monitoring, in: Business Process Management Workshops. doi:10.1007/978-3-031-82225-4_15.
- [7] Cook, N.R., 2007. Use and misuse of the receiver operating characteristic curve in risk prediction. Circulation 115, 928–935. doi:10.1161/CIRCULATIONAHA.106.672402.
- [8] Covert, I., Lundberg, S., Lee, S.I., 2020. Understanding global feature contributions with additive importance measures, in: Advances in Neural Information Processing Systems 33.
- [9] D’Amour, A., Heller, K., Moldovan, D., et al., 2022. Underspecification presents challenges for credibility in modern machine learning. Journal of Machine Learning Research 23, 1–61.

- [10] Davison, A.C., Hinkley, D.V., 1997. Bootstrap Methods and Their Application. Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, Cambridge. doi:10.1017/CB09780511802843.
- [11] Dixit, S., 2026. fieldvalue: specification surfaces for incremental predictive performance. Software, version 0.2.0. the archived release cited in the data-availability statement (DOI reserved, inserted at proof).
- [12] van Dongen, B.F., 2014. [dataset] BPI challenge 2014. 4TU.ResearchData. doi:10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35.
- [13] Fieller, E.C., 1954. Some problems in interval estimation. Journal of the Royal Statistical Society: Series B 16, 175–185. doi:10.1111/j.2517-6161.1954.tb00159.x.
- [14] Fisher, A., Rudin, C., Dominici, F., 2019. All models are wrong, but many are useful: Learning a variable’s importance by studying an entire class of prediction models simultaneously. Journal of Machine Learning Research 20, 1–81.
- [15] Ghorbani, A., Zou, J., 2019. Data Shapley: Equitable valuation of data for machine learning, in: Proceedings of the 36th International Conference on Machine Learning, pp. 2242–2251.
- [16] Holm, S., 1979. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6, 65–70.
- [17] Hutter, F., Hoos, H., Leyton-Brown, K., 2014. An efficient approach for assessing hyperparameter importance, in: Proceedings of the 31st International Conference on Machine Learning, pp. 754–762.
- [18] Jia, R., Dao, D., Wang, B., Hubis, F.A., Hynes, N., Gürel, N.M., Li, B., Zhang, C., Song, D., Spanos, C.J., 2019. Towards efficient data valuation based on the Shapley value, in: Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics, pp. 1167–1176.
- [19] Kapoor, S., Narayanan, A., 2023. Leakage and the reproducibility crisis in machine-learning-based science. Patterns 4, 100804. doi:10.1016/j.patter.2023.100804.

- [20] Kaufman, S., Rosset, S., Perlich, C., Stitelman, O., 2012. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data* 6, 15:1–15:21. doi:10.1145/2382577.2382579.
- [21] Kerr, K.F., McClelland, R.L., Brown, E.R., Lumley, T., 2011. Evaluating the incremental value of new biomarkers with integrated discrimination improvement. *American Journal of Epidemiology* 174, 364–374. doi:10.1093/aje/kwr086.
- [22] Künsch, H.R., 1989. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics* 17, 1217–1241. doi:10.1214/aos/1176347265.
- [23] Liu, C., Liu, W., Guo, N., Song, R., Gu, Y., Cheng, L., 2025. Comparative evaluation of encoding techniques for workflow process remaining time prediction for cloud applications. *Journal of Cloud Computing* 14. doi:10.1186/s13677-025-00763-8.
- [24] Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions, in: *Advances in Neural Information Processing Systems* 30, pp. 4765–4774.
- [25] Marx, C.T., Calmon, F.d.P., Ustun, B., 2020. Predictive multiplicity in classification, in: *Proceedings of the 37th International Conference on Machine Learning*, pp. 6765–6774.
- [26] Mohammed, S., Budach, L., Feuerpfeil, M., Ihde, N., Nathansen, A., Noack, N., Patzlaff, H., Naumann, F., Harmouch, H., 2025. The effects of data quality on machine learning performance on tabular data. *Information Systems* 132, 102549. doi:10.1016/j.is.2025.102549.
- [27] Patel, C.J., Burford, B., Ioannidis, J.P.A., 2015. Assessment of vibration of effects due to model specification can demonstrate the instability of observational associations. *Journal of Clinical Epidemiology* 68, 1046–1058. doi:10.1016/j.jclinepi.2015.05.029.
- [28] Pencina, M.J., D’Agostino Sr., R.B., D’Agostino Jr., R.B., Vasan, R.S., 2008. Evaluating the added predictive ability of a new marker: From area under the ROC curve to reclassification and beyond. *Statistics in Medicine* 27, 157–172. doi:10.1002/sim.2929.

- [29] Rama-Maneiro, E., Vidal, J.C., Lama, M., 2023. Deep learning for predictive business process monitoring: Review and benchmark. *IEEE Transactions on Services Computing* 16, 739–756. doi:10.1109/TSC.2021.3139807.
- [30] Romano, J.P., Wolf, M., 2005. Exact and approximate stepdown methods for multiple hypothesis testing. *Journal of the American Statistical Association* 100, 94–108. doi:10.1198/016214504000000539.
- [31] Saltelli, A., Annoni, P., Azzini, I., Campolongo, F., Ratto, M., Tarantola, S., 2010. Variance based sensitivity analysis of model output. *Computer Physics Communications* 181, 259–270. doi:10.1016/j.cpc.2009.09.018.
- [32] Semenova, L., Rudin, C., Parr, R., 2022. On the existence of simpler machine learning models. *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency* , 1827–1858doi:10.1145/3531146.3533232.
- [33] Senderovich, A., Di Francescomarino, C., Ghidini, C., Jorbina, K., Maggi, F.M., 2018. Intra and inter-case features in predictive process monitoring. *Information Systems* 74, 102–113. doi:10.1016/j.is.2017.10.005.
- [34] Silberzahn, R., Uhlmann, E.L., Martin, D.P., et al., 2018. Many analysts, one data set: Making transparent how variations in analytic choices affect results. *Advances in Methods and Practices in Psychological Science* 1, 337–356. doi:10.1177/2515245917747646.
- [35] Simonsohn, U., Simmons, J.P., Nelson, L.D., 2020. Specification curve analysis. *Nature Human Behaviour* 4, 1208–1214. doi:10.1038/s41562-020-0912-z.
- [36] Sobol’, I.M., 2001. Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates. *Mathematics and Computers in Simulation* 55, 271–280. doi:10.1016/S0378-4754(00)00270-6.
- [37] Steegen, S., Tuerlinckx, F., Gelman, A., Vanpaemel, W., 2016. Increasing transparency through a multiverse analysis. *Perspectives on Psychological Science* 11, 702–712. doi:10.1177/1745691616658637.

- [38] Steeman, W., 2013. [dataset] BPI challenge 2013, incidents. 4TU.ResearchData. doi:10.4121/uuid:500573e6-acc-4b0c-9576-aa5468b10cee.
- [39] Strobl, C., Boulesteix, A.L., Kneib, T., Augustin, T., Zeileis, A., 2008. Conditional variable importance for random forests. *BMC Bioinformatics* 9, 307. doi:10.1186/1471-2105-9-307.
- [40] Suriadi, S., Andrews, R., ter Hofstede, A.H.M., Wynn, M.T., 2017. Event log imperfection patterns for process mining. *Information Systems* 64, 132–150. doi:10.1016/j.is.2016.07.011.
- [41] Teinemaa, I., Dumas, M., La Rosa, M., Maggi, F.M., 2019. Outcome-oriented predictive process monitoring: Review and benchmark. *ACM Transactions on Knowledge Discovery from Data* 13, 17:1–17:57. doi:10.1145/3301300.
- [42] VanderWeele, T.J., Ding, P., 2017. Sensitivity analysis in observational research: Introducing the e-value. *Annals of Internal Medicine* 167, 268–274. doi:10.7326/M16-2607.
- [43] Verenich, I., Dumas, M., La Rosa, M., Maggi, F.M., Teinemaa, I., 2019. Survey and cross-benchmark comparison of remaining time prediction methods in business process monitoring. *ACM Transactions on Intelligent Systems and Technology* 10, 34:1–34:34. doi:10.1145/3331449.
- [44] Vickers, A.J., 2008. Decision analysis for the evaluation of diagnostic tests, prediction models, and molecular markers. *The American Statistician* 62, 314–320. doi:10.1198/000313008X370302.
- [45] Vickers, A.J., van Calster, B., Steyerberg, E.W., 2019. A simple, step-by-step guide to interpreting decision curve analysis. *Diagnostic and Prognostic Research* 3, 18. doi:10.1186/s41512-019-0064-7.
- [46] Vickers, A.J., Elkin, E.B., 2006. Decision curve analysis: A novel method for evaluating prediction models. *Medical Decision Making* 26, 565–574. doi:10.1177/0272989X06295361.
- [47] Vickers, A.J., Van Calster, B., Steyerberg, E.W., 2016. Net benefit approaches to the evaluation of prediction models, molecular markers, and diagnostic tests. *BMJ* 352, i6. doi:10.1136/bmj.i6.

- [48] Weytjens, H., De Weerd, J., 2022. Creating unbiased public benchmark datasets with data leakage prevention for predictive process monitoring, in: Business Process Management Workshops, pp. 18–29. doi:10.1007/978-3-030-94343-1_2.
- [49] Williamson, B.D., Gilbert, P.B., Carone, M., Simon, N., 2021. Non-parametric variable importance assessment using machine learning techniques. *Biometrics* 77, 9–22. doi:10.1111/biom.13392.